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Constructing optimal locally recoverable codes

BACHELOR'S PROJECT MATHEMATICS

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Abstract

This paper investigates constructions of optimal locally recoverable codes (LRC codes), a class of error-correcting codes that enable efficient recovery of erased codeword symbols using a small number of other symbols and achieve a Singleton-type bound specific to LRC codes. The focus is on the polynomial evaluation framework introduced by Itzhak Tamo and Alexander Barg, which generalizes Reed-Solomon codes by incorporating local recovery constraints into the encoding process. The central construction relies on evaluating structured encoding polynomials over finite fields at partitioned subsets of the field, where locality is ensured through the use of “good polynomials” — polynomials constant on each recovering set. Two algebraic methods for constructing good polynomials are developed: one based on annihilator polynomials of multiplicative or additive subgroups of finite fields, and another leveraging the interplay between additive and multiplicative structures of finite fields. In addition, we present a generalization of the original construction, and explore a systematic encoding procedure. The paper is complemented by detailed examples over small fields.

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1 Introduction

Distributed storage systems store data across multiple interconnected nodes. One of the most common types of failure in such systems is the failure of single node. Due to the design of distributed storage systems, if one node fails, the lost data can be recovered using the contents of the remaining nodes.

A simple and widely used technique in distributed storage is triple replication, where three identical copies of the data are stored across three distinct physical nodes. While this method provides strong reliability and simple recovery, it incurs a 200% storage overhead, which is becoming increasingly impractical due to the rapid growth in the volume of data that must be stored in modern society. To reduce this overhead while maintaining reliability, erasure codes are implemented.

Erasur codes work by splitting the data into k information packets and then generating $n - k$ parity packets using an encoding procedure applied to the k information packets. These n total packets are distributed across n nodes, and the original data can be recovered from any k of the n nodes [3]. This approach still achieves high reliability while significantly reducing storage overhead compared to replication. A well-known example of erasure codes used are Reed-Solomon (RS) codes, which are maximum distance separable (MDS) codes, meaning they attain the Singleton bound for minimum distance. These codes are commonly used in large-scale storage systems, such as Hadoop Distributed File System, which uses $[9, 6]$ RS codes with storage overhead of 50%, and Facebook, which has employed $[14, 10]$ RS codes with 40% storage overhead [10].

In recent years, locally recoverable codes (LRC codes) have gained considerable attention due to their ability to efficiently repair single node failures. Specifically, in an LRC code, each codeword symbol can be recovered by accessing only a small number r of other symbols, where $1 \leq r \leq k$; the parameter r is called the locality of the code. This local recovery capability addresses a key limitation of classical erasure codes such as RS code, which require access to k symbols for recovery operations, resulting in higher decoding complexity and longer recovery times.

The pioneering work of [2] established a bound on the minimum distance of an LRC code in terms of the code length n , the number k of message symbols, and the locality r . The bound in [2] resembles the classical Singleton bound and assumes linearity of the LRC code. An LRC code that achieves the bound of [2] with equality is called an optimal LRC code. Subsequent work, such as [9], generalized the results of [2] by providing a universal bound on minimum distance for both linear and nonlinear codes. For a sufficiently large finite field, [9] establishes the existence of LRC codes whose minimum distance meets the bound derived therein with equality.

A wide variety of constructions for LRC codes have been proposed. Among the most significant contributions are those of Itzhak Tamo and Alexander Barg [12], who introduced an elegant algebraic framework for constructing optimal LRC codes, that is LRC codes that attain the bound of [2]. Their approach generalizes RS codes and has inspired many follow-up works, including [7], [8], [5].

Tamo and Barg [12] present several constructions, the most general of which is known as Construction 1. This construction is based on evaluating carefully designed encoding polynomials at a specially chosen set of elements in a finite field. Central to this method is the construction of good polynomials, which are constant on specified subsets of evaluation points. The other constructions in [12] can be viewed as generalizations and extensions of Construction 1 — for example, by lifting certain divisibility assumptions on the code parameters or by extending the original local recovery framework to incorporate multiple disjoint recovering sets (i.e., availability). These extensions and generalizations have also been explored in subsequent works, such as [6], [1]. All constructions presented in [12] enable local recovery via polynomial interpolation.

This thesis focuses on a detailed study and exposition of some of the constructions of optimal LRC codes proposed in [12], particularly the theory developed in Section III. The primary objective is to present these ideas in a clear and rigorous manner, articulated independently, and supported by new, explicit examples that highlight the core mechanisms.

We begin with background material on finite fields, vector spaces over finite field, linear codes, and RS codes in Section 2, drawing on the treatments from [4] and [11]. In Section 3, we formally introduce LRC codes and their associated bounds. Section 4 is devoted to Construction 1, along with the theory of good polynomials. Building on the Tamo-Barg framework for choosing a good polynomial construction, we present a variation of their technique in which, rather than fixing the base field and varying the locality, we fix a target locality r and investigate which prime fields and their extensions support the construction of good polynomials. A generalization of Construction 1 is explored in Section 5, where a specially designed commutative algebra and a mapping from the set of encoding polynomials to a finite-dimensional vector space over a finite field are employed. Lastly, Section 6 examines a systematic variant of Construction 1, in which the original message vector appears explicitly within the constructed codeword.

2 Preliminaries

In Section 2.1, we primarily follow Sections 3 and 4 of [4]. Section 2.2 is based on material from Sections 1, 5, 6, 10, and 15 of [4], as well as Sections 2.1-2.3 of [11]. Section 2.3 draws on ideas from Sections 5.1 and 5.2 of [11].

2.1 Finite fields and vector spaces

In this paper, all codes are defined over *finite fields*. A finite field of order q is denoted by $\mathbb{F}_q$, where $q = p^n$ for some prime p and some integer $n \geq 1$. For each q , the finite field $\mathbb{F}_q$ is unique up to isomorphisms and serves as the alphabet in coding theory (see Definition 2.1).

A key structure used throughout is the *vector space* $\mathbb{F}_q^n$ over $\mathbb{F}_q$, consisting of ordered n -tuples with entries in $\mathbb{F}_q$. The total number of elements in $\mathbb{F}_q^n$ is q^n . This space is endowed with vector addition and scalar multiplication defined componentwise. That is:

1. For $\mathbf{x} = (x_0, x_1, \dots, x_{n-1})$ and $\mathbf{y} = (y_0, y_1, \dots, y_{n-1}) \in \mathbb{F}_q^n$,

$$\mathbf{x} + \mathbf{y} = (x_0 + y_0, x_1 + y_1, \dots, x_{n-1} + y_{n-1}),$$

2. For $a \in \mathbb{F}_q$ and $\mathbf{x} = (x_0, x_1, \dots, x_{n-1}) \in \mathbb{F}_q^n$,

$$a\mathbf{x} = (ax_0, ax_1, \dots, ax_{n-1}).$$

The resulting algebraic structure satisfies all vector space axioms over the field $\mathbb{F}_q$.

Example 2.1. We verify that $\mathbb{F}_2^3$, the set of all ordered triples over the binary field $\mathbb{F}_2$, is a vector space over $\mathbb{F}_2$ by checking the vector space axioms. For all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{F}_2^3$ and $a, b \in \mathbb{F}_2$:

- (i) *Closure under Addition:* $\mathbf{x} + \mathbf{y} \in \mathbb{F}_2^3$, since addition is defined componentwise and $\mathbb{F}_2$ is closed under addition.
- (ii) *Associativity of Addition:* $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$ holds componentwise by associativity of addition in $\mathbb{F}_2$.

- (iii) *Existence of Zero Vector:* The zero vector $\mathbf{0} = (0, 0, 0) \in \mathbb{F}_2^3$ and satisfies $\mathbf{x} + \mathbf{0} = \mathbf{0} + \mathbf{x} = \mathbf{x}$ componentwise
- (iv) *Existence of Additive Inverses:* For each $\mathbf{x} \in \mathbb{F}_2^3$, define $-\mathbf{x} = \mathbf{x}$, since in $\mathbb{F}_2$ we have $-x = x$. Then, $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$.
- (v) *Commutativity of Addition:* $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$, since addition in $\mathbb{F}_2$ is commutative.
- (vi) *Closure under Scalar Multiplication:* $a\mathbf{x} \in \mathbb{F}_2^3$, since scalar multiplication is defined componentwise, and $\mathbb{F}_2$ is closed under multiplication.
- (vii) *Distributivity over Vector Addition:* $a(\mathbf{x} + \mathbf{y}) = a\mathbf{x} + a\mathbf{y}$ holds componentwise by distributivity in $\mathbb{F}_2$.
- (viii) *Distributivity over Scalar Addition:* $(a + b)\mathbf{x} = a\mathbf{x} + b\mathbf{x}$, holds componentwise by distributivity in $\mathbb{F}_2$.
- (ix) *Compatibility of Scalar Multiplication:* $a(b\mathbf{x}) = (ab)\mathbf{x}$ follows since multiplication in $\mathbb{F}_2$ is associative.
- (x) *Identity Element of Scalar Multiplication:* $1\mathbf{x} = \mathbf{x}$, where 1 is the multiplicative identity in $\mathbb{F}_2$.

Since all axioms are satisfied, $\mathbb{F}_2^3$ indeed forms a vector space over $\mathbb{F}_2$.

A subspace $W \subseteq \mathbb{F}_q^n$ is a subset closed under the same operations and containing the zero vector. Any such subspace is itself a vector space and admits a basis of linearly independent vectors spanning W . Its dimension is the number of vectors in any such basis.

Example 2.2. Consider $\mathbb{F}_2$ and $\mathbb{F}_2^3$ again. Let $W \subseteq \mathbb{F}_2^3$ be the set:

$$W = \{(0, 0, 0), (1, 1, 0)\}.$$

We claim that W is a subspace of $\mathbb{F}_2^3$. To verify this, we check the subspace criteria:

1. The zero vector $(0, 0, 0) \in W$.

2. Closed under addition:

$$(1, 1, 0) + (1, 1, 0) = (0, 0, 0) \in W.$$

3. Closed under scalar multiplication:

$$0 \cdot (1, 1, 0) = (0, 0, 0), \quad 1 \cdot (1, 1, 0) = (1, 1, 0) \in W.$$

Thus, W is closed under vector addition and scalar multiplication, and it contains the zero vector. Therefore, W is a subspace of $\mathbb{F}_2^3$. Specifically, it is a 1-dimensional subspace spanned by the vector $(1, 1, 0)$.

We also make frequent use of *polynomials over finite fields*. Given a field $\mathbb{F}_q$, the set of polynomials with coefficients in $\mathbb{F}_q$ is denoted $\mathbb{F}_q[x]$. A polynomial $f(x) \in \mathbb{F}_q[x]$ has the form

$$f(x) = a_0 + a_1x + \dots + a_nx^n,$$

where $a_i \in \mathbb{F}_q$ and $n \in \mathbb{Z}_{\geq 0}$.

Two polynomials are equal if all their corresponding nonzero coefficients are equal. The *zero polynomial* is defined as the unique polynomial in which all coefficients are zero. The *degree* of a nonzero polynomial is the largest index i such that $a_i \neq 0$; by convention, the degree of the zero polynomial is $-\infty$. Polynomials of degree less than k form an $\mathbb{F}_q$ -vector space of dimension k , which we denote by $\mathbb{F}_q[x]_{\deg < k}$, and the evaluation map from this space to $\mathbb{F}_q^n$, defined by evaluating polynomials at n distinct field elements, plays a central role in code construction.

2.2 Introduction to linear codes

We introduce the foundational concepts of coding theory, focusing on block codes. These codes are essential for error detection and correction in data transmission. Among the most important classes are linear codes, which leverage the algebraic structure of vector spaces over finite fields.

2.2.1 Definitions and basic concepts

Definition 2.1 (*q*-ary code). A *q*-ary (block) code of length *n* over a finite field $\mathbb{F}_q$ is defined as a nonempty subset $C \subseteq \mathbb{F}_q^n$. Here, the field $\mathbb{F}_q$ serves as the alphabet of the code, and the elements of C are called codewords.

Note that the term “block code” emphasizes that all codewords have a fixed length *n*. Since a code is a subset of $\mathbb{F}_q^n$, which is of size q^n , the cardinality of the code can be any integer less than or equal to q^n .

Definition 2.2 (Linear code). A code $C \subseteq \mathbb{F}_q^n$ is a linear code over $\mathbb{F}_q$ if and only if C is a subspace of $\mathbb{F}_q^n$.

If C is a *k*-dimensional subspace of $\mathbb{F}_q^n$, it is called an $[n, k]$ -code, where *n* is the length of the code and *k* is its dimension. The cardinality of an $[n, k]$ -code is q^k . The quantity $n - k$ is referred to as the *redundancy* of the code.

Example 2.3. Define the set

$$C = \{(a, a + b, b) : a, b \in \mathbb{F}_3\} \subseteq \mathbb{F}_3^3.$$

We compute all possible codewords by evaluating for all $a, b \in \mathbb{F}_3$:

<i>a</i>	<i>b</i>	Codeword (<i>a</i> , <i>a</i> + <i>b</i> , <i>b</i>)
0	0	(0, 0, 0)
0	1	(0, 1, 1)
0	2	(0, 2, 2)
1	0	(1, 1, 0)
1	1	(1, 2, 1)
1	2	(1, 0, 2)
2	0	(2, 2, 0)
2	1	(2, 0, 1)
2	2	(2, 1, 2)

There are $3^2 = 9$ codewords. Clearly, the zero vector is in C . One easily verifies that for any $\mathbf{x}, \mathbf{y} \in C$ and $a \in \mathbb{F}_3$ we have $\mathbf{x} + \mathbf{y} \in C$, $a\mathbf{x} \in C$. Therefore, $C \subset \mathbb{F}_3^3$ is a 2-dimensional subspace, hence a linear $[3, 2]$ -code over $\mathbb{F}_3$.

Another example of a linear code is the repetition code.

Definition 2.3 (n-Fold repetition code). Let $\mathbb{F}_q$ be a finite field. An *n*-fold repetition code of dimension *k* over $\mathbb{F}_q$ is the linear code $C \subseteq \mathbb{F}_q^{nk}$ defined by repeating vectors of length *k*, *n* times:

$$C = \{(a_0, \dots, a_{k-1}, a_0, \dots, a_{k-1}, \dots, a_0, \dots, a_{k-1}) : (a_0, \dots, a_{k-1}) \in \mathbb{F}_q^k\} \subseteq \mathbb{F}_q^{nk}.$$

If $k = 1$, then C reduces to the standard repetition code of length *n*:

$$C_{rep} = \{(a, a, \dots, a) : a \in \mathbb{F}_q\} \subseteq \mathbb{F}_q^n,$$

that is, an $[n, 1]$ -code.

2.2.2 Distance, error detection, and error correction

To understand how codes detect and correct errors, we first need a way to measure differences between codewords. This leads us to the fundamental concept of Hamming distance.

Definition 2.4 (Hamming distance). *The Hamming distance between two vectors $\mathbf{x}$ and $\mathbf{y}$ of $\mathbb{F}_q^n$ is defined as the number of coordinates in which $\mathbf{x}$ and $\mathbf{y}$ differ. It is denoted by $d(\mathbf{x}, \mathbf{y})$. Formally, we say for $\mathbf{x} = (x_0, \dots, x_{n-1})$ and $\mathbf{y} = (y_0, \dots, y_{n-1}) \in \mathbb{F}_q^n$,*

$$d(\mathbf{x}, \mathbf{y}) = |\{i \in \{0, \dots, n-1\} : x_i \neq y_i\}|.$$

The Hamming distance is a metric. To see this we verify that it satisfies the three defining properties of a metric on the space $\mathbb{F}_q^n$.

(i) *Identity:* $d(\mathbf{x}, \mathbf{y}) = 0$ if and only if $\mathbf{x} = \mathbf{y}$.

Proof. If $\mathbf{x} = \mathbf{y}$, then $x_i = y_i$ for all i , so $d(\mathbf{x}, \mathbf{y}) = 0$. Conversely, if $d(\mathbf{x}, \mathbf{y}) = 0$, then there are no indices where $x_i \neq y_i$, hence $\mathbf{x} = \mathbf{y}$. $\square$

(ii) *Symmetry:* $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{F}_q^n$.

Proof. This holds because the definition of Hamming distance is symmetric: $x_i \neq y_i$ if and only if $y_i \neq x_i$. $\square$

(iii) *Triangle inequality:* $d(\mathbf{x}, \mathbf{y}) \leq d(\mathbf{x}, \mathbf{z}) + d(\mathbf{z}, \mathbf{y})$ for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{F}_q^n$.

Proof sketch. For each coordinate i , if $x_i \neq y_i$, then at least one of the following must be true: $x_i \neq z_i$ or $z_i \neq y_i$. Therefore, the total number of differing coordinates between $\mathbf{x}$ and $\mathbf{y}$ is at most the number of differing coordinates between $\mathbf{x}$ and $\mathbf{z}$, plus those between $\mathbf{z}$ and $\mathbf{y}$. $\square$

Example 2.4. Let $\mathbb{F}_4 = \{0, 1, \alpha, \alpha + 1\}$, where $\alpha^2 = \alpha + 1$. Consider $\mathbf{x} = (1, \alpha, 0)$, $\mathbf{y} = (1, \alpha + 1, 0) \in \mathbb{F}_4^3$. We find that $d(\mathbf{x}, \mathbf{y}) = 1$.

To further develop our tools, we introduce a related concept that measures a single vector's properties.

Definition 2.5 (Hamming weight). *The Hamming weight of a vector $\mathbf{x} \in \mathbb{F}_q^n$, denoted by $w(\mathbf{x})$, is defined as the number of nonzero components in $\mathbf{x}$.*

Example 2.5. As in Example 2.4, take $\mathbb{F}_4 = \{0, 1, \alpha, \alpha + 1\}$, $\mathbf{x} = (1, \alpha, 0)$, $\mathbf{y} = (1, \alpha + 1, 0) \in \mathbb{F}_4^3$. Then $w(\mathbf{x}) = w(\mathbf{y}) = 2$.

Lemma 2.6. *For any $\mathbf{x}, \mathbf{y} \in \mathbb{F}_q^n$, the Hamming distance between them satisfies:*

$$d(\mathbf{x}, \mathbf{y}) = w(\mathbf{x} - \mathbf{y}),$$

where $w(\cdot)$ denotes the Hamming weight.

Proof. The nonzero entries of the vector $\mathbf{x} - \mathbf{y}$ correspond precisely to the coordinates where $\mathbf{x}$ and $\mathbf{y}$ differ. $\square$

Example 2.6. As in Example 2.4, take $\mathbb{F}_4 = \{0, 1, \alpha, \alpha + 1\}$, $\mathbf{x} = (1, \alpha, 0)$, $\mathbf{y} = (1, \alpha + 1, 0) \in \mathbb{F}_4^3$. Observe that

$$w(\mathbf{x} - \mathbf{y}) = w(0, \alpha - (\alpha + 1), 0) = w(0, -1, 0) = 1 = d(\mathbf{x}, \mathbf{y}).$$

Definition 2.7 (Minimum distance). *The minimum distance of a code $C \subseteq \mathbb{F}_q^n$, denoted $d(C)$, is the smallest Hamming distance between any two distinct codewords in C , formally given by:*

$$d(C) = \min_{\mathbf{x}, \mathbf{y} \in C: \mathbf{x} \neq \mathbf{y}} d(\mathbf{x}, \mathbf{y})$$

A linear $[n, k]$ -code can also be written as an $[n, k, d]$ -code, where d is the minimum distance of the code.

Theorem 2.8. For a linear $[n, k, d]$ -code C over $\mathbb{F}_q$, the minimum distance is equal to the smallest Hamming weight among all nonzero codewords in C , i.e.,

$$d(C) = \min_{\mathbf{c} \in C \setminus \{0\}} w(\mathbf{c}).$$

Proof. Since C is linear we have that if $\mathbf{c}_1, \mathbf{c}_2 \in C$, then $\mathbf{c}_1 - \mathbf{c}_2 \in C$. By Lemma 2.6, $d(\mathbf{c}_1, \mathbf{c}_2) = w(\mathbf{c}_1 - \mathbf{c}_2)$ thus

$$d(C) = \min_{\mathbf{c}_1, \mathbf{c}_2 \in C: \mathbf{c}_1 \neq \mathbf{c}_2} d(\mathbf{c}_1, \mathbf{c}_2) = \min_{\mathbf{c}_1, \mathbf{c}_2 \in C: \mathbf{c}_1 \neq \mathbf{c}_2} w(\mathbf{c}_1 - \mathbf{c}_2) = \min_{\mathbf{c} \in C \setminus \{0\}} w(\mathbf{c}).$$

□

The minimum distance is fundamental in determining the error detection and correction capability of a code, as it reflects how distinguishable the codewords are from one another.

Theorem 2.9 (Error detection and correction). Let C be a code with minimum distance $d(C)$.

- (i) If $d(C) \geq s + 1$, then C can detect up to s errors in any codeword.
- (ii) If $d(C) \geq 2t + 1$, then C can correct up to t errors in any codeword.

Proof. (i) Assume that the minimum distance of the code satisfies $d(C) \geq s + 1$. Suppose a codeword $\mathbf{x} \in C$ is transmitted, and during transmission, at most s errors occur. That is, the received word $\mathbf{y} \in \mathbb{F}_q^n$ differs from $\mathbf{x}$ in at most s positions, so $d(\mathbf{x}, \mathbf{y}) \leq s$.

Suppose $\mathbf{y} \in C$. Then $\mathbf{y} = \mathbf{x}$, since any other codeword must be at least $d(C) \geq s + 1$ positions away from $\mathbf{x}$, which contradicts $d(\mathbf{x}, \mathbf{y}) \leq s$. Thus, if an error occurred and changed any symbols, the received word cannot lie in C , i.e., cannot be a different codeword, and the error is therefore detected.

- (ii) Assume $d(C) \geq 2t + 1$. Suppose a codeword $\mathbf{x} \in C$ is transmitted, and at most t errors occur during transmission. Then the received word $\mathbf{y} \in \mathbb{F}_q^n$ satisfies $d(\mathbf{x}, \mathbf{y}) \leq t$.

Suppose, for contradiction, that there exists another codeword $\mathbf{x}' \in C$, with $\mathbf{x} \neq \mathbf{x}'$, such that $\mathbf{x}'$ is also within distance t of $\mathbf{y}$: i.e., $d(\mathbf{x}', \mathbf{y}) \leq t$.

Then, by the triangle inequality,

$$d(\mathbf{x}, \mathbf{x}') \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{x}') \leq t + t = 2t,$$

which contradicts that assumptions that $d(C) \geq 2t + 1$. Therefore, no such $\mathbf{x}'$ can exist and $\mathbf{x}$ is the unique codeword closest to $\mathbf{y}$.

□

Corollary 2.10. Let C be a code with minimum distance $d(C)$. Then:

- (i) It can detect up to $d(C) - 1$ errors,
- (ii) And can correct up to $\left\lfloor \frac{d(C)-1}{2} \right\rfloor$, where $\lfloor \cdot \rfloor$ is the floor function.

Proof. (i) From Theorem 2.9 (part (i)), we can detect up to s errors provided that $d(C) \geq s + 1$. The largest possible s is $d(C) - 1$, i.e.,

$$d(C) \geq s + 1 \iff d(C) - 1 \geq s.$$

- (ii) From Theorem 2.9 (part (ii)), we can correct up to t errors provided that $d(C) \geq 2t + 1$. Solving for t , we obtain $t \leq \frac{d(C)-1}{2}$. Since t must be an integer, we take

$$t = \left\lfloor \frac{d(C) - 1}{2} \right\rfloor.$$

□

2.2.3 Rate and Singleton bound

To evaluate code performance, we need to introduce some quantitative measures. We begin with the concept of a rate.

Definition 2.11 (Rate). *For an $[n, k]$ -code C , the rate $R(C)$ is defined as*

$$R(C) = \frac{k}{n} \in [0, 1].$$

The rate measures the ratio of information symbols to the total symbols in a codeword, and thus reflects the efficiency of the code. A low rate (close to 0) indicates a highly redundant code, which requires a lot of energy to transmit a short message. In contrast, a rate close to 1 corresponds to an efficient code, where the proportion of information symbols in each codeword is close to 1.

One of the most important concepts in coding theory is the Singleton bound that reflects the relationship between a code's parameters.

Definition 2.12 (Singleton bound). *Let C be an $[n, k, d]$ -code, and denote $d := d(C)$. Then*

$$d \leq n - k + 1 \tag{1}$$

is called the Singleton bound.

The Singleton bound establishes a fundamental limit on the minimum distance d of a code, based on its length n and dimension k . It captures the inherent trade-off between these parameters and provides a theoretical upper bound on the error-correcting capability of any code with specified length and dimension.

Definition 2.13 (MDS code). *A linear $[n, k, d]$ -code over $\mathbb{F}_q$ is said to be a maximum distance separable (MDS) code, if it attains the Singleton bound, namely, if*

$$d = n - k + 1.$$

MDS codes are optimal in the sense that they offer the greatest possible error or erasure correction capability for the given parameters n and k .

2.2.4 Encoding and decoding

The algebraic structure of linear codes enable efficient encoding and decoding procedures.

Definition 2.14 (Generator matrix). *Let C be a linear $[n, k, d]$ -code over $\mathbb{F}_q$. A generator matrix G of C is a $k \times n$ matrix whose rows form a basis of C . The rank of G is equal to the dimension k of C .*

It is important to note that generator matrices for a given linear code are generally not unique as different bases yield different generator matrices.

Encoding of linear codes

Let C be a linear $[n, k, d]$ -code over $\mathbb{F}_q$. Then C is a k -dimensional subspace of $\mathbb{F}_q^n$. The *encoding* process maps a message vector $\mathbf{u} \in \mathbb{F}_q^k$ to a codeword $\mathbf{c} \in C$, via

$$\mathbb{F}_q^k \rightarrow C, \quad \mathbf{u} \mapsto \mathbf{u}G,$$

where G is a generator matrix.

Observe that this mapping is bijective. Indeed, since $\text{rank}(G) = k$, the equation $\mathbf{u}G = \mathbf{0}$ has only the trivial solution $\mathbf{u} = \mathbf{0}$. This is because the rows of G are linearly independent, so no nonzero linear combination of them can yield the zero vector. Hence, the kernel is trivial, which implies injectivity. Furthermore, since the domain and codomain have the same dimension — and hence the same cardinality — surjectivity follows.

Decoding of linear codes

Decoding is the process of recovering the original message $\mathbf{u} \in \mathbb{F}_q^k$ from a (possibly corrupted) received word $\mathbf{x} \in \mathbb{F}_q^n$.

2.3 Reed-Solomon codes

Reed–Solomon (RS) codes are a powerful family of linear block codes that are widely used in real-world applications due to their exceptional error-correcting performance and elegant algebraic structure. They are constructed via evaluation of low-degree polynomials over finite fields and are known to attain the Singleton bound, making them examples of MDS codes.

Definition 2.15 (Reed-Solomon code). *Consider the finite field $\mathbb{F}_q$, and let $\alpha_1, \dots, \alpha_n \in \mathbb{F}_q$ be distinct elements. Fix a positive integer $k \leq n$. An $[n, k, d]$ Reed-Solomon (RS) code C_{RS} over $\mathbb{F}_q$ is defined as*

$$C_{RS} := RS_k(\alpha_1, \dots, \alpha_n) = \{ (m(\alpha_1), m(\alpha_2), \dots, m(\alpha_n)) \in \mathbb{F}_q^n : m(x) \in \mathbb{F}_q[x], \deg(m(x)) < k \},$$

where $m(x)$ is the polynomial

$$m(x) = \sum_{i=0}^{k-1} m_i x^i = m_0 + m_1 x + \dots + m_{k-1} x^{k-1} \in \mathbb{F}_q[x]_{\deg < k}$$

associated with a message vector $\mathbf{m} = (m_0, m_1, \dots, m_{k-1}) \in \mathbb{F}_q^k$.

In other words, an RS code consists of all evaluation vectors of polynomials of degree less than k , derived from message vectors of length k and evaluated at the fixed distinct points $\alpha_1, \dots, \alpha_n \in \mathbb{F}_q$, known as *code locators*.

This construction yields a linear $[n, k, d]$ -code over $\mathbb{F}_q$, since the set of polynomials of degree less than k forms a k -dimensional vector space over $\mathbb{F}_q$, and the evaluation map

$$\text{ev} : \mathbb{F}_q[x]_{\deg < k} \rightarrow \mathbb{F}_q^n, \quad f \mapsto (f(\alpha_1), \dots, f(\alpha_n))$$

is linear and injective. Specifically, for $f(x), g(x) \in \mathbb{F}_q[x]_{\deg < k}$ and $\lambda \in \mathbb{F}_q$, we have

$$\begin{aligned} \text{ev}(f + \lambda g) &= ((f + \lambda g)(\alpha_1), \dots, (f + \lambda g)(\alpha_n)) = ((f(\alpha_1) + \lambda g(\alpha_1), \dots, (f(\alpha_n) + \lambda g(\alpha_n))) \\ &= \text{ev}(f) + \lambda \text{ev}(g), \end{aligned}$$

so the map is linear. To see that ev is injective, suppose $\text{ev}(f) = \text{ev}(g)$. Then $f(\alpha_i) = g(\alpha_i)$ for

all $i = 1, \dots, n$, so the difference $h(x) := f(x) - g(x) \in \mathbb{F}_q[x]_{\deg < k}$ satisfies

$$h(\alpha_1) = \dots = h(\alpha_n) = 0.$$

Since $\deg(h) < k$, the polynomial $h(x)$ can have at most $\deg(h) < n$ distinct roots unless it is identically zero. Consequently, as h vanishes at n distinct points, it must be the zero polynomial. Hence $f = g$, and ev is injective.

An example of the construction is given below.

Example 2.7. Consider $\mathbb{F}_5 = \{0, 1, 2, 3, 4\}$, and the RS code $RS_2(1, 2, 3, 4) \subseteq \mathbb{F}_5^4$. Each codeword is obtained by evaluating a polynomial $m(x) \in \mathbb{F}_5[x]$ of degree less than 2 at $\alpha_1 = 1, \alpha_2 = 2, \alpha_3 = 3, \alpha_4 = 4$.

Let the message vector be $\mathbf{m} = (2, 3) \in \mathbb{F}_5^2$ and consider $m(x) = 2 + 3x \in \mathbb{F}_5[x]_{\deg < 2}$. Then the codeword is:

$$(m(1), m(2), m(3), m(4)) = (0, 3, 1, 4).$$

Thus $(0, 3, 1, 4) \in RS_2(1, 2, 3, 4)$.

While it is technically possible to choose $n \leq q$, we typically assume $n \leq q - 1$. This ensures that all code locators are nonzero elements of the field, i.e., members of the multiplicative group $\mathbb{F}_q^\times$. Although this restriction is not essential for many of the fundamental properties of RS codes, certain decoding algorithms rely on the assumption that each code locator has a multiplicative inverse in $\mathbb{F}_q$. Additionally, a given RS code can often be defined using multiple distinct sets of code locators.

In particular, when $n \mid q - 1$, the cyclic group $(\mathbb{F}_q)^\times$ of order $q - 1$ contains an element α of multiplicative order n . This allows for a natural and uniform choice of code locators

$$\alpha_j = \alpha^{j-1}, \quad 1 \leq j \leq n.$$

That is,

$$\{\alpha_1, \alpha_2, \dots, \alpha_n\} = \{1, \alpha, \dots, \alpha^{n-1}\}.$$

Remark 2.16. The RS code can also be described via the map

$$\mathbb{F}_q^k \rightarrow C_{RS}, \quad \mathbf{m} \mapsto \mathbf{m}G_{RS}.$$

where

$$G_{RS} = \begin{pmatrix} 1 & 1 & \dots & 1 \\ \alpha_1 & \alpha_2 & \dots & \alpha_n \\ \alpha_1^2 & \alpha_2^2 & \dots & \alpha_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_1^{k-1} & \alpha_2^{k-1} & \dots & \alpha_n^{k-1} \end{pmatrix} \in \mathbb{F}_q^{k \times n}$$

is a generator matrix. In the case of RS codes, this generator matrix is a Vandermonde matrix, where the rows correspond to monomials — that is, the i^{th} row of G_{RS} corresponds to the monomial x^i — and the columns correspond to the evaluation points; specifically, the j^{th} column corresponds to the evaluation at α_j . It follows that

$$C_{RS} = \{\mathbf{m}G_{RS} : \mathbf{m} \in \mathbb{F}_q^k\}.$$

The Vandermonde matrix G_{RS} generates the codewords via polynomial evaluation, and has full rank provided that the α_j are distinct.

RS codes are maximum distance separable codes, as stated in the following proposition.

Proposition 2.17. Every $[n, k, d]$ RS code over $\mathbb{F}_q$ satisfies the Singleton bound with equality:

$$d = n - k + 1.$$

Hence, RS codes are MDS codes.

Proof sketch. Let $f \in \mathbb{F}_q[x]_{\deg < k}$ be a nonzero polynomial of degree less than k . Such a polynomial has at most $k-1$ roots in $\mathbb{F}_q$. Thus, in any nonzero codeword $(f(\alpha_1), \dots, f(\alpha_n))$, there are at most $k-1$ zero coordinates. This implies that the Hamming weight of every nonzero codeword is at least $n - k + 1$, and hence by Lemma 2.6 the minimum distance $d \geq n - k + 1$. Combining this with the Singleton bound (1) yields equality. $\square$

Erasure recovery in RS codes

When a component of a codeword $\mathbf{c} \in C_{RS}$ is erased, it can be recovered using any k unerased components of $\mathbf{c}$. This is achieved by first reconstructing the original message polynomial of degree less than k via polynomial interpolation through k known evaluation points. Once the message polynomial is recovered, the missing component is obtained by evaluating this polynomial at the corresponding code locator.

3 Locally recoverable codes

In this section, we formally introduce locally recoverable codes and present two fundamental bounds associated with them: a special rate bound and a special minimum distance bound.

A code $C \subseteq \mathbb{F}_q^n$ over a finite field $\mathbb{F}_q$ is called a *locally recoverable code* (LRC code) if, for any erased coordinate i , the component c_i of a codeword $\mathbf{c} \in C$ can be recovered using a subset of (at most) r other components in $\mathbf{c}$. We call the parameter r the *locality* of the code. That is, we say that C has locality r if, for each coordinate $i \in \{1, \dots, n\}$, the component c_i is a function of (at most) r other components of the same codeword. More precisely, for every $i \in \{1, \dots, n\}$, there exists a subset $I_i \subseteq \{1, \dots, n\} \setminus \{i\}$ with $|I_i| \leq r$ and a function

$$\phi_i : \mathbb{F}_q^{|I_i|} \rightarrow \mathbb{F}_q$$

such that for every codewords $\mathbf{c} \in C$, we have

$$c_i = \phi_i((c_j)_{j \in I_i}).$$

Letting $I := \{1, \dots, n\}$, we give a more formal definition.

Definition 3.1 (Locality). Fix $a \in \mathbb{F}_q$, and consider a set of codewords

$$C(i, a) = \{\mathbf{c} \in C : c_i = a\}, \quad i \in I.$$

The code C has locality r if, for every $i \in I$, there exists a subset $I_i \subseteq I \setminus \{i\}$ with $|I_i| \leq r$ such that the sets $C(i, a)$ restricted to the coordinates in I_i are disjoint for distinct values of a ; that is,

$$C_{I_i}(i, a) \cap C_{I_i}(i, a') = \emptyset \quad \text{for all } a \neq a'. \quad (2)$$

The set I_i is called the *recovering set* of the coordinate i . The condition in (2) implies that the values of a codeword on the coordinates I_i uniquely determine the value at coordinate i . In other words, no two codewords that differ at position i can agree on all the positions in I_i , thus the value at i is locally determined by I_i . Since $|I_i| \leq r$, one only needs to access at most r other components of the codeword to determine the i^{th} component. This is the essence of locality r . We use (n, k, r) to refer to an LRC code of length n , cardinality q^k , and locality r .

- Example 3.1.** (i) A 2-fold repetition code of length $2k$, in which each component is repeated twice, has locality $r = 1$. In this case, any component can be recovered by accessing its duplicate.
- (ii) An $[n, k]$ MDS code (e.g., a Reed-Solomon code) has locality $r = k$, since recovering a single erased component requires accessing k other components.

Since an (n, k, r) LRC code uses k information symbols to produce codewords of length n , any codeword can be reconstructed by accessing any k components excluding the erased one. Therefore, the locality parameter r must satisfy $1 \leq r \leq k$. Smaller values of r are preferable, as they enable more efficient local recovery.

In designing codes, two key objectives are to achieve a large minimum distance and a high code rate. We present two bounds on the parameters of an (n, k, r) LRC code.

Theorem 3.2. Let $C \subseteq \mathbb{F}_q^n$ be a (n, k, r) LRC code. The rate R of the C satisfies

$$R = \frac{k}{n} \leq \frac{r}{r+1}, \quad (3)$$

and the minimum distance d of C satisfies

$$d \leq n - k - \left\lceil \frac{k}{r} \right\rceil + 2. \quad (4)$$

Proofs of these upper bounds can be found in [2], [9], and [12]. The latter assumes that the (n, k, r) LRC code is linear, and resembles the classical Singleton bound. A code that achieves equality in the Singleton-type bound (4) is called an *optimal* LRC code.

Remark 3.3. When $r = k$, the bound in (4) reduces to the classical Singleton bound:

$$d \leq n - k + 1.$$

Thus, MDS codes such as Reed-Solomon codes represent a special case of optimal LRC codes with locality $r = k$.

On the other hand, when $r = 1$, the bound in (4) simplifies to

$$d \leq n - 2k + 2 = 2 \left(\frac{n}{2} - k + 1 \right)$$

By duplicating each component in an $(\frac{n}{2}, k)$ MDS code, one obtains an optimal LRC code with locality $r = 1$.

In the remainder of the paper, the primary focus will be on constructing LRC codes with locality in the range $1 < r < k$.

4 Construction 1 — optimal linear (n, k, r) LRC codes

In this section, we present the first explicit construction of optimal linear (n, k, r) LRC codes over a finite field $\mathbb{F}_q$ introduced by Tamo and Barg. This method, referred to as **Construction 1**, is a general polynomial evaluation based approach that extends Reed-Solomon codes while incorporating local recovery constraints. We also explore algebraic constructions of good polynomials that exploit the additive and multiplicative structures of finite fields. In addition, we introduce a variation on the technique proposed by Tamo and Barg for selecting appropriate good polynomial constructions. The content of this section is based on Sections III.A and III.B of [12].

4.1 Overview

The key idea of **Construction 1** is to evaluate specially designed encoding polynomials at a carefully chosen, algebraically structured set of points in $\mathbb{F}_q$. Specifically, the evaluation set is partitioned into disjoint subsets of size $r + 1$, and the construction relies on the existence of a specially designed polynomial $g(x) \in \mathbb{F}_q[x]$ — called a *good polynomial* — which is constant on each subset of the partition. The encoding polynomials are then built from powers of $g(x)$, with coefficients being low-degree polynomials, allowing the code to possess both good distance properties and local recovery.

Construction 1 assumes the following structural conditions on the parameters:

1. $q \geq n$, so that sufficiently many distinct evaluation points are available in the field;
2. $r \mid k$, to allow for a uniform structure in the encoding polynomial;
3. $(r + 1) \mid n$, to ensure a clean partition into recovering sets.

The first assumption is a general requirement for the construction of LRC codes. As we will see later in the paper, the second assumption — namely, that $r \mid k$ — can be lifted without affecting the validity of the construction. We also note that the final assumption, $r + 1 \mid n$, can likewise be removed; see, for instance, [12, Section V.A].

4.2 Construction 1: General polynomial evaluation construction

In Section 4.2.1, we describe the setup for **Construction 1**. The construction itself is detailed in Section 4.2.2. In Section 4.2.3, we discuss the local recovery procedure, which we emphasize constitutes a general approach applicable to all LRC code constructions presented in [12]. Finally, in Section 4.2.3, we prove that the codes obtained via **Construction 1** are indeed locally recoverable and that they attain the optimal minimum distance bound for LRC codes, given by (4).

4.2.1 Partition and good polynomial

We begin with the two key ingredients required for the construction: a carefully chosen partition of the evaluation set and a good polynomial.

Let $\mathcal{A}$ be a partition of a set $A \subseteq \mathbb{F}_q$ of size n into $m = \frac{n}{r+1}$ disjoint sets, each of size $r + 1$:

$$\mathcal{A} = \{A_1, A_2, \dots, A_m\} \quad \text{such that } A = \bigsqcup_{l=1}^m A_l \text{ with } |A_l| = r + 1 \text{ for all } l = 1, \dots, m.$$

The entire set A will serve as our evaluation set, that is, the set of points at which we evaluate our encoding polynomial.

A polynomial $g(x) \in \mathbb{F}_q[x]$ is said to be *good* with respect to this partition if:

- $\deg(g) = r + 1$,
- $g(\alpha) = g(\beta)$ for all $\alpha, \beta \in A_l$, for each $l = 1, \dots, m$, i.e., g is constant on each A_l .

Such a polynomial enables the construction of encoding polynomials with locality constraints built in.

4.2.2 Construction 1: Encoding polynomial structure

Let $n \leq q$ be the target code length and assume that the conditions in Section 4.1 hold. Moreover, as described in Section 4.2.1, let $\mathcal{A}$ be a partition of some set $A \subseteq \mathbb{F}_q$ consisting of n distinct field elements and assume that a good polynomial $g(x)$ exists.

Let the message vector be $\mathbf{a} = (a_{i,j}) \in \mathbb{F}_q^k$, indexed by $i = 0, \dots, r-1$ and $j = 0, \dots, \frac{k}{r}-1$, so that $r \cdot \frac{k}{r} = k$, and thus $\mathbf{a}$ indeed lies in $\mathbb{F}_q^k$. For each $i = 0, \dots, r-1$, define the *coefficient polynomials*:

$$f_i(x) = \sum_{j=0}^{\frac{k}{r}-1} a_{i,j} g(x)^j, \quad (5)$$

which satisfy $\deg(f_i) \leq (\frac{k}{r} - 1) \deg(g) = (\frac{k}{r} - 1)(r + 1)$. Then define *encoding polynomial*:

$$f_{\mathbf{a}}(x) = \sum_{i=0}^{r-1} f_i(x) x^i. \quad (6)$$

The corresponding codeword is the evaluation vector of $f_{\mathbf{a}}$ at all the points in A :

$$(f_{\mathbf{a}}(\alpha_1), f_{\mathbf{a}}(\alpha_2), \dots, f_{\mathbf{a}}(\alpha_n)), \quad \text{where } \alpha_i \in A \text{ and } i \neq j.$$

The entire (n, k, r) LRC code $C \subseteq \mathbb{F}_q^n$ is defined as the set of all such n -dimensional evaluation vectors:

$$C = \{ (f_{\mathbf{a}}(\alpha))_{\alpha \in A} : \mathbf{a} \in \mathbb{F}_q^k \}. \quad (7)$$

We refer to each element $\alpha \in A$ as a *location* and each corresponding component $f_{\mathbf{a}}(\alpha)$ as a *symbol* of the codeword.

Remark 4.1. The code C constructed in (7) is linear. To see this, let $\lambda \in \mathbb{F}_q$, and let $\mathbf{a}, \mathbf{b} \in \mathbb{F}_q^k$ be two nonzero message vectors with corresponding encoding polynomials $f_{\mathbf{a}}(x), f_{\mathbf{b}}(x) \in \mathbb{F}_q[x]$, defined as follows:

$$\begin{aligned} f_{\mathbf{a}}(x) &= \sum_{i=0}^{r-1} f_i(x) x^i, \quad \text{where } f_i(x) = \sum_{j=0}^{\frac{k}{r}-1} a_{i,j} g(x)^j, \\ f_{\mathbf{b}}(x) &= \sum_{i=0}^{r-1} \tilde{f}_i(x) x^i, \quad \text{where } \tilde{f}_i(x) = \sum_{j=0}^{\frac{k}{r}-1} b_{i,j} g(x)^j. \end{aligned}$$

We verify that for any two evaluation vectors $(f_{\mathbf{a}}(\alpha))_{\alpha \in A}$ and $(f_{\mathbf{b}}(\alpha))_{\alpha \in A}$ in C ,

- $(f_{\mathbf{a}}(\alpha))_{\alpha \in A} + (f_{\mathbf{b}}(\alpha))_{\alpha \in A} \in C$ and
- $\lambda (f_{\mathbf{a}}(\alpha))_{\alpha \in A} \in C$.

To do this, we show that the encoding polynomial in (6) is linear for a given location $\alpha \in A$:

$$\begin{aligned} f_{\mathbf{a}}(\alpha) + f_{\mathbf{b}}(\alpha) &= \sum_{i=0}^{r-1} f_i(\alpha) \alpha^i + \sum_{i=0}^{r-1} \tilde{f}_i(\alpha) \alpha^i = \sum_{i=0}^{r-1} \alpha^i (f_i(\alpha) + \tilde{f}_i(\alpha)) = \sum_{i=0}^{r-1} \alpha^i \sum_{j=0}^{\frac{k}{r}-1} (a_{i,j} g(\alpha)^j + b_{i,j} g(\alpha)^j) \\ &= \sum_{i=0}^{r-1} \alpha^i \sum_{j=0}^{\frac{k}{r}-1} g(\alpha)^j (a_{i,j} + b_{i,j}) \\ &= f_{\mathbf{a}+\mathbf{b}}(\alpha), \end{aligned}$$

thus $(f_{\mathbf{a}}(\alpha))_{\alpha \in A} + (f_{\mathbf{b}}(\alpha))_{\alpha \in A} \in C$. Moreover,

$$\lambda f_{\mathbf{a}}(\alpha) = \lambda \sum_{i=0}^{r-1} f_i(\alpha) \alpha^i = \sum_{i=0}^{r-1} \lambda f_i(\alpha) \alpha^i = \sum_{i=0}^{r-1} \alpha^i \sum_{j=0}^{\frac{k}{r}-1} (\lambda \cdot a_{i,j}) g(\alpha)^j = f_{\lambda \mathbf{a}}(\alpha),$$

and hence $\lambda(f_{\mathbf{a}}(\alpha))_{\alpha \in A} \in C$.

We note that two linear codes C and C' are said to be equivalent if their codewords differ only by a fixed permutation of symbols. That is, C and C' represent the same code up to reordering of symbols within each codeword. The fundamental code parameters — such as length, dimension, and locality — are invariant under permutations of the evaluation sequence. These parameters depend only on the set of evaluation points and the encoding polynomial.

In this paper, we adopt a specific evaluation order for implementation: we first evaluate $f_{\mathbf{a}}(x)$ at all locations in the subset A_1 of our partition $\mathcal{A} = \{A_1, \dots, A_m\}$, then at the locations in A_2 , and so on, ending with those in A_m .

Crucially, local recovery — presented in the next section — does not rely on a symbol's position in the codeword, but rather on the recovering set associated with its corresponding location. This means the relative ordering of evaluating points, whether within or across recovering sets, does not affect the code's local recovery capability.

In Theorem 4.2, we will see that the codes constructed in this way are indeed locally recoverable, have dimension k , and are distance-optimal.

We present an example that illustrates the construction process.

Example 4.1. We aim to construct an (n, k, r) LRC code over the finite field $\mathbb{F}_q$ via **Construction 1**. Recall that we require $r \mid k$ and $(r+1) \mid n$, where $n \leq q$.

Let us choose the parameters $(n, k, r) = (12, 4, 2)$. Indeed, $2 \mid 4$ and $3 \mid 12$, so the divisibility conditions are satisfied. We must now pick a suitable finite field. Since we need $n = 12 \leq q$, let us pick $q = 19$. This choice will become more clear later. As a flash forward, we will see that we need to ensure the existence of a subgroup of $\mathbb{F}_{19}^\times$ of order $r+1$ in order to obtain a valid partition. Since $3 \mid (19-1) = 18$, the field $\mathbb{F}_{19}$ works.

We need $\frac{n}{r+1} = \frac{12}{3} = 4$ disjoint sets of size 3. Let

$$\mathcal{A} = \left\{ A_1 = \{1, 7, 11\}, A_2 = \{2, 14, 3\}, A_3 = \{4, 9, 6\}, A_4 = \{5, 16, 17\} \right\}.$$

We will later describe a systematic method to select the disjoint sets in the partition. Namely, we will see that A_1 is in fact a subgroup of order 3 in $\mathbb{F}_{19}^\times$ and A_2, A_3, A_4 are its cosets:

$$A_2 = 2A_1, \quad A_3 = 4A_1, \quad A_4 = 5A_1.$$

In order to proceed, we need a good polynomial $g(x) \in \mathbb{F}_{19}[x]$ of degree $r+1 = 3$ that is constant on each A_i . For this example, we simply state the polynomial without justification (its construction will be explained later). We take $g(x) = x^3$.

Since $r-1 = 1$ and $\frac{k}{r} - 1 = 1$, let the message vector be

$$\mathbf{a} = (a_{0,0}, a_{0,1}, a_{1,0}, a_{1,1}) = (1, 1, 1, 1) \in \mathbb{F}_{19}^4.$$

Define the encoding polynomial

$$\begin{aligned}
f_{\mathbf{a}}(x) &= \underbrace{(a_{0,0} + a_{0,1}g(x))}_{f_0} + \underbrace{(a_{1,0} + a_{1,1}g(x))}_{f_1} x \\
&= (a_{0,0} + a_{0,1}x^3) + (a_{1,0} + a_{1,1}x^3)x \\
&= a_{0,0} + a_{1,0}x + a_{0,1}x^3 + a_{1,1}x^4 \\
&= 1 + x + x^3 + x^4.
\end{aligned}$$

To obtain the corresponding codeword, we evaluate $f_{\mathbf{a}}(x)$ at all 12 locations in $A = A_1 \cup A_2 \cup A_3 \cup A_4$, taken in the order

$$(1, 7, 11, 2, 14, 3, 4, 9, 6, 5, 16, 17).$$

The final codeword is:

$$\begin{aligned}
\mathbf{c} &= (f_{\mathbf{a}}(\alpha)_{\alpha \in A}) = (f_{\mathbf{a}}(1), f_{\mathbf{a}}(7), f_{\mathbf{a}}(11), f_{\mathbf{a}}(2), f_{\mathbf{a}}(14), f_{\mathbf{a}}(3), f_{\mathbf{a}}(4), f_{\mathbf{a}}(9), f_{\mathbf{a}}(6), f_{\mathbf{a}}(5), f_{\mathbf{a}}(16), f_{\mathbf{a}}(17)) \\
&= (4, 16, 5, 8, 2, 17, 2, 4, 18, 15, 14, 7).
\end{aligned}$$

We note that a $(12, 4, 2)$ LRC code can also be constructed over $\mathbb{F}_{13}$, for instance. This is the smallest possible field in which such a construction can be carried out, since we require $n \leq q$.

4.2.3 Local recovery via interpolation

In this section we present the recovery method that is central to all the code constructions in [12], in particular **Construction 1**. It allows for the recovery of any single erased symbol using only a small, fixed number r of other symbols, thereby illustrating the principle of locality.

Suppose we have an erasure at the symbol $f_{\mathbf{a}}(\alpha)$ corresponding to location $\alpha \in A_l$, where A_l is one of the subsets in the partition $\mathcal{A}$. Denote the erased symbol as $c_{\alpha} := f_{\mathbf{a}}(\alpha)$. Recall that $|A_l| = r + 1$. For each of the r remaining locations in $A_l \setminus \{\alpha\}$, let c_{β} denote the symbol corresponding to $\beta \in A_l \setminus \{\alpha\}$.

Given that each $\beta \in A_l \setminus \{\alpha\}$ is distinct, we can find a unique Lagrange interpolating polynomial $\delta(x)$ of degree at most $r - 1$ that passes through these r known points c_{β} . Specifically, we have

$$\delta(x) = \sum_{\beta \in A_l \setminus \{\alpha\}} c_{\beta} \prod_{\beta' \in A_l \setminus \{\alpha, \beta\}} \frac{x - \beta'}{\beta - \beta'} \quad (8)$$

with $\delta(\beta) = c_{\beta}$ for every $\beta \in A_l \setminus \{\alpha\}$.

In order to recover the missing symbol c_{α} , we evaluate $\delta(x)$ at α , i.e., we set $c_{\alpha} = \delta(\alpha)$. We refer to $\delta(x)$ as the *decoding polynomial* for the symbol c_{α} , and the subset $A_l \setminus \{\alpha\}$ of size r serves as the recovering set for c_{α} .

Since recovering an erased symbol involves the construction of an interpolating polynomial for the r remaining locations in its recovering set through the corresponding r symbols, we see that any erased symbol can be recovered using r other symbols, yielding locality r .

In the proof of Theorem 4.2 we demonstrate that codes constructed via **Construction 1** indeed guarantee local recovery: any missing symbol can be uniquely reconstructed by performing polynomial interpolation of the r other symbols corresponding to its recovering set.

Namely, we will see that the correctness of this procedure follows from the algebraic structure of the

code. In **Construction 1** we define the encoding polynomial $f_{\mathbf{a}}(x) \in \mathbb{F}_q[x]$ as

$$f_{\mathbf{a}}(x) = \sum_{i=0}^{r-1} f_i(x)x^i.$$

where each coefficient polynomial f_i is a linear combination of powers of a fixed good polynomial:

$$f_i(x) = \sum_{j=0}^{\frac{k}{r}-1} a_{i,j}g(x)^j.$$

The key point is that the good polynomial $g(x)$ is chosen so that it is constant on each subset $A_l \subset A$. It follows that for each fixed $i = 0, \dots, r-1$, the coefficient polynomial restricted to A_l is a polynomial on A_l , i.e.,

$$f_i|_{A_l} = c_i.$$

Therefore, the restriction of the encoding polynomial $f_{\mathbf{a}}(x)$ to A_l becomes a

$$f_{\mathbf{a}}(x)|_{A_l} = \sum_{i=0}^{r-1} c_i x^i,$$

which is a polynomial of degree at most $r-1$.

The symbols $f_{\mathbf{a}}(\alpha)$ for $\alpha \in A_l$ lie on this polynomial of degree at most $r-1$. This means that if any one value $f_{\mathbf{a}}(\alpha)$ for $\alpha \in A_l$ is erased, it can be recovered by interpolating the degree- $(r-1)$ polynomial $f_{\mathbf{a}}(x)|_{A_l}$ from the remaining r symbols of the locations in $A_l \setminus \{\alpha\}$, and then evaluating $f_{\mathbf{a}}(x)|_{A_l}$ at the location α [6]. This requires only accessing r other symbols hence achieving locality r .

Example 4.1 (Continued). Suppose $f_{\mathbf{a}}(1)$ is erased. Since $1 \in A_1 = \{1, 7, 11\}$ and $|A_1 \setminus \{1\}| = 2$, the value $f_{\mathbf{a}}(1)$ can, by construction, be recovered using the two $(r=2)$ codeword symbols $f_{\mathbf{a}}(7)$ and $f_{\mathbf{a}}(11)$ at locations 7 and 11. We use Lagrange interpolation from (8) to recover the missing value, working in $\mathbb{F}_{19}$:

$$\begin{aligned} \delta(x) &= f_{\mathbf{a}}(7) \left(\frac{x-11}{7-11} \right) + f_{\mathbf{a}}(11) \left(\frac{x-7}{11-7} \right) = 16 \left(\frac{x-11}{-4} \right) + 5 \left(\frac{x-7}{4} \right) \\ &= 16 \left(\frac{x-11}{15} \right) + 5 \left(\frac{x-7}{4} \right) \\ &= 16(x-11)(14) + 5(x-7)(5) \\ &= 15x - 13 + 6x - 4 \\ &= 2x - 17 \\ &= 2x + 2, \end{aligned}$$

Evaluating this at $x = 1$, we correctly get

$$\delta(1) = 2 + 2 = 4.$$

4.2.4 Linearity and optimality of Construction 1

In this section, we formally prove that **Construction 1** is valid with respect to the recovery procedure, has dimension k , and attains the Singleton-type bound for LRC codes, thus ensuring optimality.

Theorem 4.2. The linear code C defined in (7) via **Construction 1** has dimension k and is an optimal (n, k, r) LRC code. That is, C has locality r and attains the Singleton-type bound (4) for LRC codes.

Proof. 1. *Locality property:* To demonstrate the locality property of the code, consider a subset $A_l \in \mathcal{A}$ from the partition of the evaluation set. Suppose the symbol $c_\alpha := f_{\mathbf{a}}(\alpha)$ at location $\alpha \in A_l$ is lost and needs to be recovered.

We define the *decoding polynomial* (8), as follows:

$$\partial_\alpha(x) = \sum_{i=0}^{r-1} f_i(\alpha)x^i, \quad (9)$$

where the $f_i(x)$ are the coefficient polynomials (5).

We show that (8) and (9) are the same (recall that the decoding polynomial must be unique). Each $f_i(x)$ is a linear combination of powers of the good polynomial $g(x)$, and hence remains constant on each set A_l . That is, for any $\beta \in A_l$, we have

$$f_i(\beta) = f_i(\alpha), \quad \text{for all } i = 0, \dots, r-1. \quad (10)$$

Using this property and (9), it follows that for any $\beta \in A_l$,

$$\partial_\alpha(\beta) = \sum_{i=0}^{r-1} f_i(\alpha)\beta^i = \sum_{i=0}^{r-1} f_i(\beta)\beta^i = f_{\mathbf{a}}(\beta).$$

In other words, $\partial_\alpha(x)$ agrees with $f_{\mathbf{a}}(x)$ on the entire set A_l .

Since $\partial_\alpha(x)$ is a polynomial of degree at most $r-1$, it can be interpolated from r symbols corresponding to the remaining r distinct locations in $A_l \setminus \{\alpha\}$, which agrees with (8).

Once $\partial_\alpha(x)$ is constructed via (9), the erased symbol is recovered as $\partial_\alpha(\alpha)$. Thus, the locality property is satisfied: the erased code symbol c_α can be recovered by accessing only the r other symbols c_β for $\beta \in A_j \setminus \{\alpha\}$.

2. *Dimension:* For indices $i = 0, \dots, r-1$ and $j = 0, \dots, \frac{k}{r}-1$, the total number of distinct polynomials $g(x)^j x^i$ in the expression for $f_{\mathbf{a}}(x)$ is $r \cdot \frac{k}{r} = k$. The k polynomials $g(x)^j x^i$ are constructed such that each polynomial has a distinct degree. This ensures that they are linearly independent over the finite field $\mathbb{F}_q$. Consequently, the mapping $\mathbf{a} \mapsto f_{\mathbf{a}}(x)$ is injective.

The degree of each coefficient polynomial f_i in (6) is at most $(\frac{k}{r}-1)(r+1)$, so the total degree of the encoding polynomial $f_{\mathbf{a}}$ is at most

$$(\frac{k}{r}-1)(r+1) + (r-1) = k + \frac{k}{r} - 2,$$

i.e.,

$$\deg(f_{\mathbf{a}}) \leq k + \frac{k}{r} - 2.$$

Using (3) we then get:

$$k + \frac{k}{r} - 2 \leq \frac{nr}{r+1} + \frac{n}{r+1} - 2 = n - 2,$$

hence,

$$\deg(f_{\mathbf{a}}) \leq n - 2.$$

Since distinct encoding polynomials $f_{\mathbf{a}}$ and $f_{\mathbf{b}}$ have degree at most $n-2$, their difference $f_{\mathbf{a}} - f_{\mathbf{b}}$ is

a nonzero polynomial of degree $\leq n - 2$. Consequently, the difference polynomial can have at most $n - 2$ distinct roots. However, the evaluation set A contains n distinct points, so the evaluation vector $((f_{\mathbf{a}} - f_{\mathbf{b}})(\alpha))_{\alpha \in A}$ cannot be the zero vector. Therefore codewords corresponding to distinct encoding polynomials are distinct and there are exactly q^k unique codewords. In other words, distinct messages map to distinct codewords (i.e., the map $\mathbb{F}_q^k \rightarrow \mathbb{F}_q^n$ is injective), and thus the dimension of the code is k .

3. *Minimum distance:* To prove that the code is distance-optimal, we leverage the linearity of the code. Recall that the minimum distance of a linear code equals the minimum Hamming weight Theorem 2.8. Observe that the codeword $\mathbf{c}_{\mathbf{a}} \in \mathbb{F}_q^n$, corresponding to a nonzero message vector $\mathbf{a} \in \mathbb{F}_q^k$, has at most $\deg(f_{\mathbf{a}})$ zeros among the n evaluation points. Thus, the number of nonzero entries of $\mathbf{c}_{\mathbf{a}}$ is at least $n - \deg(f_{\mathbf{a}})$, that is

$$w(\mathbf{c}_{\mathbf{a}}) \geq n - \deg(f_{\mathbf{a}}).$$

It follows that the minimum distance of the code satisfies:

$$d(C) \geq n - \max_{f_{\mathbf{a}}, \mathbf{a} \in \mathbb{F}_q^k} \deg(f_{\mathbf{a}}) = n - k - \frac{k}{r} + 2.$$

When this lower bound is combined with the Singleton-type bound (4), we obtain equality. $\square$

It follows from the proof (part (i)) that since $\partial_{\alpha}(\alpha) = f_{\mathbf{a}}(\alpha)$ for all $\alpha \in A_l$, $\delta(x)$ in (8) satisfies $\delta(\alpha) = f_{\mathbf{a}}(\alpha)$ for all $\alpha \in A_l$ (as we showed that $\partial_{\alpha}(x) = \delta(x)$). In other words, $\delta(x)$ depends only on the index l of the recovering set A_l , and not on the specific location $\alpha \in A_l$. In particular, any two locations $\alpha_1, \alpha_2 \in A_l$ share the same decoding polynomial $\delta(x)$, confirming that the recovery procedure is uniform across all elements of a given recovering set.

Example 4.1 (Continued). *We found the encoding polynomial*

$$f_{\mathbf{a}}(x) = a_{0,0} + a_{1,0}x + a_{0,1}x^3 + a_{1,1}x^4.$$

Since the number of terms in $f_{\mathbf{a}}$ is four — in other words, the number of polynomials of the form $g(x)^j x^i$ appearing in $f_{\mathbf{a}}$ is four — it follows from the proof of Theorem 4.2 (part(2)) that the dimension of the code is indeed $k = 4$.

Furthermore, we have $\deg(f_{\mathbf{a}}) \leq 4$ for any $\mathbf{a} \in \mathbb{F}_q^k$, thus:

$$d(C) \geq 12 - 4 = 8.$$

By the Singleton-type bound for LRC codes (4):

$$d(C) \leq 12 - 4 - 2 + 2 = 8,$$

so we conclude that $d = 8$, and therefore the $(12, 4, 2)$ LRC code we constructed is indeed optimal.

We end this section with a key observation concerning the dimension parameter k .

Remark 4.3. *As shown in the proof of Theorem 4.2, the dimension of the code depends solely on the length of the message vector. This is because the map from message vectors $\mathbf{a} \in \mathbb{F}_q^k$ to encoding polynomials $f_{\mathbf{a}}(x)$ is injective, and the subsequent map from encoding polynomials to codewords $\mathbf{c} \in \mathbb{F}_q^n$ is also injective. Therefore, the overall composition $\mathbb{F}_q^k \rightarrow \mathbb{F}_q^n$ is injective.*

As we will demonstrate in Section 4.4, the divisibility condition $r \mid k$ mentioned in Section 4.2.1, initially assumed necessary for **Construction 1**, can, in fact, be relaxed. This has an important implication: to determine the admissible values of the dimension k for a desired (n, k, r) LRC code, it suffices to ensure injectivity of the overall encoding map.

To this end, for fixed parameters n and r , and one can apply the rate bound given in (3),

$$\frac{k}{n} \leq \frac{r}{r+1}$$

which yields the upper bound

$$k \leq \frac{nr}{r+1}. \quad (11)$$

This inequality indicates that any message vector in $\mathbb{F}_q^k$ with $k \leq \frac{nr}{r+1}$ can be used to encode an (n, k, r) LRC code.

It follows that, for any fixed n and r , an (n, k, r) LRC code can be constructed for every dimension k satisfying (11), provided that a suitable field, an appropriate partition of an evaluation set, and good polynomial are available.

4.3 Algebraic constructions of good polynomials

In order to implement **Construction 1** and ensure that the resulting code satisfies the locality property, we require a good polynomial $g(x) \in \mathbb{F}_q[x]$ of degree $r+1$ that is constant on each set in a partition $\mathcal{A} = \{A_1, \dots, A_m\}$ of the evaluation set $A \subseteq \mathbb{F}_q$. This section provides two frameworks for constructing such polynomials by exploiting group-theoretic structures within finite fields. The two good polynomial constructions simultaneously yield an appropriate partition.

4.3.1 Option 1: Annihilator polynomial

Consider the finite field $\mathbb{F}_q$, where $q = p^n$. The multiplicative group $\mathbb{F}_q^\times$ of $\mathbb{F}_q$ is a cyclic group of order $q-1$. On the other hand, the additive group $\mathbb{F}_q^+$ is isomorphic to $(\mathbb{Z}/p\mathbb{Z})^n$, i.e., a direct product of n copies of the additive group of $\mathbb{Z}/p\mathbb{Z}$. That is, $\mathbb{F}_q^+ \cong (\mathbb{Z}/p\mathbb{Z})^n \cong \mathbb{F}_p^n$ (isomorphic as additive groups), so $\mathbb{F}_q$ is a n -dimensional vector space over its prime field $\mathbb{F}_p \cong \mathbb{Z}/p\mathbb{Z}$.

Let H be a subgroup of either the multiplicative group $\mathbb{F}_q^\times$ or the additive group $\mathbb{F}_q^+$. Then, by Lagrange's theorem, the order of H divides the order $\mathbb{F}_q^\times$ (or $\mathbb{F}_q^+$), and H partitions the ambient group into disjoint cosets.

In the multiplicative case, where $H \leq \mathbb{F}_q^\times$, there are

$$[\mathbb{F}_q^\times : H] = \frac{q-1}{|H|}.$$

distinct cosets, each of the form $aH = \{ah : h \in H\}$ for some $a \in \mathbb{F}_q^\times$. For any two elements $a, b \in \mathbb{F}_q^\times$ we have $aH = bH$ precisely when $ab^{-1} \in H$. In other words, a and b lie in the same coset if and only if $ab^{-1} \in H$. Otherwise, $aH \cap bH = \emptyset$.

In the additive case, where $H \leq \mathbb{F}_q^+$, there are

$$[\mathbb{F}_q^+ : H] = \frac{q}{|H|}.$$

distinct cosets, each of form $a + H = \{a + h : h \in H\}$ for some $a \in \mathbb{F}_q$. Two elements $a, b \in \mathbb{F}_q$ lie in the

same coset if and only if $a - b \in H$. Since in the additive case the order of H is p^t for some nonnegative integer t , we have $\frac{q}{|H|} = p^{n-t}$.

We can construct a good polynomial via the annihilator polynomial of a subgroup H , which vanishes on all elements of H :

Proposition 4.4. *Let H be a subgroup of either $\mathbb{F}_q^\times$ or $\mathbb{F}_q^+$. Define the annihilator polynomial of H as*

$$g(x) = \prod_{h \in H} (x - h). \quad (12)$$

Then $g(x)$ is constant on each coset of H in both the multiplicative and additive settings.

Proof. 1. *Multiplicative case:* Let $H \leq \mathbb{F}_q^\times$, and consider an arbitrary coset $aH = \{ah : h \in H\}$. Suppose $\bar{h} \in H$ so that $a, a\bar{h} \in aH$. Then

$$\begin{aligned} g(a\bar{h}) &= \prod_{h \in H} (a\bar{h} - h) = \prod_{h \in H} \bar{h}(a - h\bar{h}^{-1}) = \bar{h}^{|H|} \prod_{h \in H} (a - h\bar{h}^{-1}) \\ &= \prod_{h \in H} (a - h\bar{h}^{-1}) \\ &= \prod_{h \in H} (a - h) \\ &= g(a), \end{aligned}$$

where in the fourth equality we use that $\bar{h} \in H$ and H is a group of order $|H|$, so $\bar{h}^{|H|} = 1$ (by Lagrange's theorem, since the group operation is multiplication), and in the fifth equality we use that multiplication by $\bar{h}^{-1}$ defines a permutation of H , so the map $h \mapsto h\bar{h}^{-1}$ is a bijection on H .

2. *Additive case:* Let $H \leq \mathbb{F}_q^+$, and consider the coset $a + H = \{a + h : h \in H\}$. Suppose $\bar{h} \in H$ so that $a, a + \bar{h} \in a + H$. Then

$$\begin{aligned} g(a + \bar{h}) &= \prod_{h \in H} ((a + \bar{h}) - h) = \prod_{h \in H} (a - (h - \bar{h})) \\ &= \prod_{h \in H} (a - h) \\ &= g(a), \end{aligned}$$

where in the third equality we use that $h \mapsto h - \bar{h}$ is a bijection of H onto itself so the set $\{h - \bar{h} : h \in H\}$ is a reordering of H .

In both cases, the product is independent of the choice of $\bar{h}$, hence g is constant on each coset of H . $\square$

In the multiplicative case, note that since H is a cyclic group of order $|H|$, its elements satisfy $h^{|H|} = 1$, and thus the annihilator polynomial in (12) can be written as

$$g(x) = x^{|H|} - 1,$$

since both expression vanish exactly on the elements of H . In other words, the multiplicative subgroup H is precisely the set of all roots of $x^{|H|} - 1$ in $\mathbb{F}_q$.

Let $\gamma_1, \dots, \gamma_m \in \mathbb{F}_q^\times$ be representatives of these m distinct cosets of H . Define $A_i = \gamma_i H = \{\gamma_i h : h \in H\}$. Then for $x = \gamma_i h \in A_i$,

$$g(x) = (\gamma_i h)^{|H|} - 1 = \gamma_i^{|H|} h^{|H|} - 1 = \gamma_i^{|H|} - 1,$$

since $h \in H$ and $h^{|H|} = 1$.

Hence, the polynomial $g(x) = x^{|H|} - 1$ takes the same value on every element of a coset $A_i = \gamma_i H$, and takes distinct values on different cosets (provided $\gamma_i^{|H|} \neq \gamma_j^{|H|}$).

Remark 4.5. *The polynomial $g(x) = x^{|H|}$ is often preferable in practice. This is because, when analyzing the behaviour of $g(x)$ on cosets of a multiplicative subgroup H , it behaves identically to $x^{|H|} - 1$: both are constant on each coset of H . These two polynomials differ only by an additive constant, which does not affect the fact that they are both constant on each coset. More formally, the sets $\{1, x^{|H|} - 1\}$ and $\{1, x^{|H|}\}$ span the same 2-dimensional vector space over $\mathbb{F}_q$:*

$$x^{|H|} - 1 \in \text{span}(1, x^{|H|})$$

and

$$x^{|H|} = (x^{|H|} - 1) + 1 \in \text{span}(1, x^{|H|} - 1).$$

Since the two spans are equal, and LRC codes are defined by evaluating functions from such a span, the resulting codes are equivalent — they have the same length, dimension, and minimum distance.

Annihilator polynomials associated with subgroups $\mathbb{F}_q^\times$ or $\mathbb{F}_q^+$ constitute a valuable class of good polynomials. When the evaluation set $A \subseteq \mathbb{F}_q$ is chosen as a union of disjoint cosets of a subgroup H , the partition $\mathcal{A}$ consists of these cosets. In this setup, the length n of the resulting code is a multiple of the size of each coset. To achieve locality r , we take H to be of order $r + 1$, hence n is a multiple of $r + 1$ constrained by $n \leq q - 1$ in the multiplicative case and $n \leq q$ in the additive case. Since H is a subgroup, its order divides the order of the full group, and hence $q \bmod (r + 1)$ equals either 1 or 0, depending on the group structure we are picking.

Before presenting concrete examples, we briefly summarize the above. Suppose that our target code length is n , and let us fix a multiplicative subgroup $H \leq \mathbb{F}_q^\times$ of order $r + 1$. Since $(r + 1) \mid (q - 1)$, the group $\mathbb{F}_q^\times$ is partitioned into $\frac{q-1}{r+1}$ disjoint cosets of H . To construct our code, we select $m \leq \frac{q-1}{r+1}$ such cosets to form the evaluation set. Each coset has cardinality $r + 1$, so the total number of evaluation points is

$$n = m(r + 1).$$

We then take $g(x) = x^{r+1}$ as a good polynomial.

In the additive case, the construction proceeds analogously, but with $m \leq \frac{q}{r+1}$ cosets and with the annihilator polynomial in (12) as the good polynomial.

We see that the choice of whether to use a multiplicative or additive subgroup depends on the structure of the finite field $\mathbb{F}_q$. For example, if $q = p$ for a prime p , then $\mathbb{F}_q^+$ has no proper additive subgroups, and so a multiplicative subgroup must be used. Consequently, the parameters of LRC codes constructed using subgroups are inherently constrained by the availability and sizes of those subgroups within a chosen field.

Example 4.1 (Revisited). *Since 19 is prime, we use the multiplicative subgroup $H = \{1, 7, 11\} \leq \mathbb{F}_{19}^\times$ and its cosets, along with the good polynomial $g(x) = x^3$, which is derived from the annihilator polynomial $g(x) = x^3 - 1$. We did not use two of the total number of cosets because only 12 evaluation points were needed.*

Example 4.2. *In this example, we construct an optimal $(9, 4, 2)$ LRC code using the additive group of the field $\mathbb{F}_9$. Note that in this case $n = q$, and the divisibility conditions from Section 4.2.1 hold.*

Let $\mathbb{F}_9 = \mathbb{F}_3[\alpha]$, where $\alpha^2 + 1 = 0$, so that $\mathbb{F}_9 = \{a + b\alpha : a, b \in \mathbb{F}_3\}$. Consider the additive subgroup

$$H = \{0, \alpha, 2\alpha\},$$

of order $r + 1 = 3$. There are $[\mathbb{F}_9 : H] = \frac{9}{3} = 3$ additive cosets of H in $\mathbb{F}_9$, which form the partition

$$\mathcal{A} = \left\{ \begin{array}{lll} A_1 & = H & = \{0, \alpha, 2\alpha\}, \\ A_2 & = 1 + H & = \{1, 1 + \alpha, 1 + 2\alpha\}, \\ A_3 & = 2 + H & = \{2, 2 + \alpha, 2 + 2\alpha\} \end{array} \right\}$$

of the evaluation set A of size 9.

We construct the good polynomial $g(x)$ using (12):

$$g(x) = \prod_{h \in H} (x - h) = x(x + \alpha)(x - \alpha) = x(x^2 - \alpha^2) = x(x^2 + 1) = x^3 + x.$$

For completeness, we verify that $g(x)$ is constant on each A_i in the partition:

- By definition (12), $g(x) = 0$ on A_1 :

$$g(0) = 0 + 0 = 0,$$

$$g(\alpha) = \alpha^3 + \alpha = \alpha^2\alpha + \alpha = -\alpha + \alpha = 0,$$

$$g(2\alpha) = (2\alpha)^3 + (2\alpha) = 8\alpha^3 + 2\alpha = -2\alpha + 2\alpha = 0.$$

- On A_2 , $g(x) = 2$:

$$g(1) = 1 + 1 = 2,$$

$$\begin{aligned} g(1 + \alpha) &= (1 + \alpha)^3 + (1 + \alpha) = (1 + 3\alpha + 3\alpha^2 + \alpha^3) + (1 + \alpha) \\ &= (1 - \alpha) + (1 + \alpha) = 2, \end{aligned}$$

$$\begin{aligned} g(1 + 2\alpha) &= (1 + 2\alpha)^3 + (1 + 2\alpha) = (1 + 6\alpha + 12\alpha^2 + 8\alpha^3) + (1 + 2\alpha) \\ &= (1 - 2\alpha) + (1 + 2\alpha) = 2. \end{aligned}$$

- On A_3 , $g(x) = 1$:

$$g(2) = 8 + 2 = 1,$$

$$\begin{aligned} g(2 + \alpha) &= (2 + \alpha)^3 + (2 + \alpha) = (8 + 12\alpha + 6\alpha^2 + \alpha^3) + (2 + \alpha) \\ &= (2 - \alpha) + (2 + \alpha) = 1, \end{aligned}$$

$$\begin{aligned} g(2 + 2\alpha) &= (2 + 2\alpha)^3 + (2 + 2\alpha) = (8 + 24\alpha + 24\alpha^2 + 8\alpha^3) + (2 + 2\alpha) \\ &= (2 - 2\alpha) + (2 + 2\alpha) = 1. \end{aligned}$$

We have $r - 1 = 1$ and $\frac{k}{r} - 1 = 1$. Let the message be $\mathbf{a} = (a_{i,j})$, with $i, j \in \{0, 1\}$. For each i define the coefficient polynomial:

$$f_i(x) = a_{i,0} + a_{i,1}g(x).$$

The encoding polynomial is then:

$$f_{\mathbf{a}}(x) = f_0(x) + f_1(x)x.$$

We evaluate $f_{\mathbf{a}}(x)$ at all 9 locations in $A = A_1 \cup A_2 \cup A_3$. By Theorem 4.2, the resulting code has dimension $k = 4$, and is an optimal LRC code. That is,

$$d = n - k - \left\lceil \frac{k}{r} \right\rceil + 2 = 9 - 4 - 2 + 2 = 5.$$

In comparison with a $(9, 4)$ MDS code — which achieves a minimum distance of 6 via the Singleton bound — this construction reduces the minimum distance to 5, but achieves a twofold improvement in locality, decreasing it from 4 to 2.

4.3.2 Option 2: Combined construction

The additive and multiplicative structures of a finite field can be simultaneously exploited to develop a more general method for constructing good polynomials.

Before presenting the main theorem that underpins this method, we consider two preliminary notions. Given two subsets $H, G \subset \mathbb{F}_q$, we say that H is *closed under multiplication by G* if $gH \subseteq H$ for all $g \in G$. Moreover, the following lemma characterizes the number of m^{th} roots of unity in a finite field.

Lemma 4.6. *Let $\mathbb{F}_q$ be a finite field, and let $m \geq 1$ be an integer. Then the number of distinct m^{th} roots of unity in $\mathbb{F}_q$ is given by*

$$|\{x \in \mathbb{F}_q^\times : x^m = 1\}| = \gcd(m, q - 1).$$

Proof. The multiplicative group $\mathbb{F}_q^\times$ is cyclic of order $q - 1$. Let γ be a generator of $\mathbb{F}_q^\times$. Then every $x \in \mathbb{F}_q^\times$ can be written as $x = \gamma^i$ for some $0 \leq i < q - 1$.

An element $x = \gamma^i$ satisfies $x^m = 1$ if and only if

$$(\gamma^i)^m = \gamma^{im} = 1.$$

Since γ has order $q - 1$, this occurs if and only if $(q - 1)$ divides im , i.e.,

$$(q - 1) \mid im.$$

This is equivalent to the condition

$$\frac{(q - 1)}{\gcd(m, q - 1)} \mid i.$$

Hence, $x = \gamma^i$ is an m^{th} root of unity in $\mathbb{F}_q$ if and only if i is a multiple of $\frac{q-1}{\gcd(m, q-1)}$ in the range $0 \leq i < q - 1$. In other words, the m^{th} roots of unity in $\mathbb{F}_q$ are precisely the elements

$$\gamma^{k \cdot \frac{q-1}{\gcd(m, q-1)}} \quad \text{for } k = 0, 1, \dots, \gcd(m, q - 1) - 1.$$

In particular, there are exactly $\gcd(m, q - 1)$ such elements, completing the proof. $\square$

It follows that if $m \mid q - 1$, then $\gcd(m, q - 1) = m$, and $\mathbb{F}_q^\times$ contains a unique cyclic subgroup of order m , consisting precisely of all the m^{th} roots of unity in $\mathbb{F}_q$, generated by a primitive m^{th} root of unity.

We now state the main theorem. This is followed by some remarks and subsequently the proof.

Theorem 4.7. *Let $\ell, s, m \in \mathbb{Z}$ such that $\ell \mid s$, and suppose p is a prime with $p^\ell \equiv 1 \pmod{m}$. Let H be an additive subgroup of the finite field $\mathbb{F}_{p^s}$, and suppose that H is closed under multiplication by $\mathbb{F}_{p^\ell}$. Let $\alpha_1, \dots, \alpha_m$ denote the m^{th} roots of unity in $\mathbb{F}_{p^s}$.*

Define the polynomial

$$g(x) = \prod_{i=1}^m \prod_{h \in H} (x + h + \alpha_i). \tag{13}$$

Then for any $b \in \mathbb{F}_{p^s}$, the polynomial $g(x)$ is constant on the union of cosets of H ,

$$\bigcup_{1 \leq i \leq m} H + b\alpha_i.$$

Moreover, the cardinality of this union is given by

$$\left| \bigcup_{1 \leq i \leq m} H + b\alpha_i \right| = \begin{cases} |H| & \text{if } b \in H, \\ m|H| & \text{if } b \notin H. \end{cases}$$

In essence, the theorem works by partitioning by the finite field $\mathbb{F}_{p^s}$ into structured sets via a subgroup $H \leq \mathbb{F}_{p^s}$, thereby distinguishing between elements that lie inside H and those that lie outside.

We make two observations. First, since $\ell \mid s$, it follows that $\mathbb{F}_{p^\ell} \subseteq \mathbb{F}_{p^s}$. Under the assumption that $p^\ell = 1 \pmod m$, we have $\gcd(p^\ell - 1, m) = m$, and hence $\mathbb{F}_{p^\ell}$ contains exactly m distinct m^{th} roots of unity. Since $\mathbb{F}_{p^\ell} \subseteq \mathbb{F}_{p^s}$, these roots also lie in $\mathbb{F}_{p^s}$. Furthermore, since $x^m - 1$ splits completely over $\mathbb{F}_{p^\ell}$, all possible m^{th} roots of unity are already contained in $\mathbb{F}_{p^\ell}$, and thus also in $\mathbb{F}_{p^s}$. That is, no additional m^{th} roots can appear in any larger extension field (in this case, $\mathbb{F}_{p^s}$).

Second, since $\ell \mid s$, the field $\mathbb{F}_{p^s}$ can be viewed as a vector space of dimension $\frac{s}{\ell}$ over the subfield $\mathbb{F}_{p^\ell}$, closed under multiplication by $\mathbb{F}_{p^\ell}$. Any $\mathbb{F}_{p^\ell}$ -subspace of $\mathbb{F}_{p^s}$ is an additive subgroup closed under multiplication by $\mathbb{F}_{p^\ell}$ (i.e., it is closed under scalar multiplication). Therefore, in the context of the theorem, the subgroup $H \leq \mathbb{F}_{p^s}$ can be taken to be any $\mathbb{F}_{p^\ell}$ -subspace of dimension $1 \leq t \leq s/\ell$. The size of such a subgroup H is

$$|H| = (p^\ell)^t = p^{t\ell}.$$

Proof. Let $\bar{h} + b\alpha_j$ be an arbitrary element of the coset union, where $\bar{h} \in H$, α_j is an m^{th} root of unity, and $b \in \mathbb{F}_{p^s}$. Then

$$g(\bar{h} + b\alpha_j) = \prod_{i=1}^m \prod_{h \in H} (\bar{h} + b\alpha_j + h + \alpha_i).$$

Since $\bar{h} + h \in H$, we may rewrite this as

$$g(\bar{h} + b\alpha_j) = \prod_{i=1}^m \prod_{\tilde{h} \in H} (b\alpha_j + \tilde{h} + \alpha_i).$$

Next, we factor out α_j :

$$g(\bar{h} + b\alpha_j) = \alpha_j^{-m|H|} \prod_{i=1}^m \prod_{\tilde{h} \in H} (b + \tilde{h}\alpha_j^{-1} + \alpha_i\alpha_j^{-1}) = \prod_{i=1}^m \prod_{\tilde{h} \in H} (b + \tilde{h}\alpha_j^{-1} + \alpha_i\alpha_j^{-1})$$

since $\alpha_j^m = 1$. Because H is closed under multiplication by $\mathbb{F}_{p^\ell}$ (and in particular under multiplication by α_j^{-1}), we have $\tilde{h}\alpha_j^{-1} \in H$. Moreover, $\alpha_i\alpha_j^{-1}$ is another m^{th} root of unity. Thus, the product simplifies to

$$g(\bar{h} + b\alpha_j) = \prod_{k=1}^m \prod_{h' \in H} (b + h' + \alpha_k) = g(b).$$

We have therefore shown, using change of variables and the assumption that H is closed under multiplication by any m^{th} root of unity (since it is closed under multiplication by $\mathbb{F}_{p^\ell}$), that g takes the same value for all elements in the coset union $\bigcup_{1 \leq i \leq m} H + b\alpha_i$.

To compute the size of this coset union, consider two distinct m^{th} roots of unity $\alpha_i \neq \alpha_j$, and sup-

pose $H + b\alpha_i = H + b\alpha_j$. Then

$$b\alpha_i - b\alpha_j \in H \iff b(\alpha_i - \alpha_j) \in H.$$

Since $\alpha_i \neq \alpha_j$, the difference $\alpha_i - \alpha_j$ is a nonzero element of $\mathbb{F}_{p^\ell}$. As H is closed under multiplication by $\mathbb{F}_{p^\ell}$, the condition $b(\alpha_i - \alpha_j) \in H$ implies $b \in H$. Therefore, if $b \in H$, then $H + b\alpha_i = H$ for all i (because $b\alpha_i \in H$), and the union is simply H , of size $|H|$. Otherwise, if $b \notin H$, the m cosets $H + b\alpha_i$ are pairwise disjoint, and the union has size $m|H|$. $\square$

Observe that the polynomial $g(x)$ in (13) is defined as a product over m roots of unity and over all elements $h \in H$. Thus, the degree of $g(x)$ is $m|H|$. This implies that the equation $g(x) = c$, where c is a constant, can have at most $m|H|$ distinct roots. This property ensures that $g(x)$ takes distinct values on distinct union of cosets

$$U_b = \bigcup_{i=1}^m H + b\alpha_i.$$

Specifically, for any $b \notin H$, the union U_b has $m|H|$ distinct elements, and $g(x)$ takes a constant value $g(b)$ for all of them (as shown in the proof above). Since $g(x) - g(b)$ vanishes on all $m|H|$ points of U_b , and $\deg(g) = m|H|$, the polynomial cannot vanish elsewhere. Therefore, the value $g(b)$ is uniquely associated with U_b for $b \notin H$. In the case where $b \in H$, we have $U_b = H$, and $g(x)$ takes a distinct value $g(b)$ on H , which differs from all $g(b')$ where $b' \notin H$.

This distinctness of values implies that $g(x)$ partitions $\mathbb{F}_{p^s}$ into:

- One subset of size $|H|$ (when $b \in H$),
- $\frac{p^s - |H|}{m|H|}$ subsets of size $m|H|$ (for $b \notin H$).

This partition arises because the $p^s - |H|$ elements outside H are grouped into disjoint sets U_b each of size $m|H|$, on which $g(x)$ takes distinct values. Consequently, this partitioning facilitates the construction of optimal LRC codes over the field $\mathbb{F}_{p^s}$ via **Construction 1** provided the code length $n \leq p^s$ is chosen such that $m|H|$ divides n .

Recall that we found $|H| = p^{t\ell}$, where $1 \leq t \leq \frac{s}{\ell}$. Thus, we may write $m|H| = mp^{t'\ell}$, where $t' = t\ell$, i.e., t' is a multiple of ℓ . Theorem 4.7 therefore enables the construction of good polynomials that are constant on sets of size $mp^{t'}$, thereby enabling the construction of optimal LRC codes via **Construction 1** over a field of characteristic p .

In fact, if one aims to construct an LRC code over a field of a specific characteristic p , then the theorem provides a powerful and flexible framework for constructing good polynomials over a wide range of parameter choices. As discussed, our objective is to construct a good polynomial that is constant on sets of size $mp^{t'}$. To this end, we require only that the prime p and the integer m satisfy $p \nmid m$. This guarantees the existence of m^{th} roots of unity in $\mathbb{F}_{p^\ell}$, since a primitive m^{th} root of unity exists in $\mathbb{F}_{p^\ell}$ if and only if $m \mid p^\ell - 1$. If instead $p \mid m$, then $m \nmid p^\ell - 1$ because $p \mid p^\ell - 1$. Moreover, choosing ℓ to be the smallest integer such that $p^\ell \equiv 1 \pmod{m}$ ensures that $\mathbb{F}_{p^\ell}$ is the minimal field extension of $\mathbb{F}_p$ containing a primitive m^{th} root of unity. A smaller value of ℓ offers greater flexibility in the choice of $\mathbb{F}_{p^s}$ and t' , thus accommodating a broader range of code lengths n as well as dimensions k by Remark 4.3. Note that by Euler's Theorem, if $\gcd(m, p) = 1$, then $p^{\phi(m)} \equiv 1 \pmod{m}$, where $\phi(\cdot)$ denotes Euler's totient function. It follows that $\ell \leq \phi(m)$ (and in particular, $\ell \mid \phi(m)$).

Example 4.3. Let us fix the characteristic $p = 5$, and choose code parameters $n = 20$ and $r = 9$. In order to construct an optimal LRC code via **Construction 1**, we must identify a suitable finite extension $\mathbb{F}_{5^s}$ of $\mathbb{F}_5$, and construct a good polynomial $g(x)$ that is constant on $\frac{n}{r+1} = \frac{20}{10} = 2$ disjoint subsets

of size $r + 1 = 10$ in a partition of an evaluation set $A \subseteq \mathbb{F}_{5^s}$ of size 20.

Note that a subset of size $r + 1$ can be interpreted as being of size $mp^{t'}$, where m is an integer coprime to the characteristic p . In this case, observe that $10 = 2 \cdot 5$, so $m = 2$ and $p^{t'} = 5$, so $t' = 1$. Moreover, $\ell = 1$ is the smallest integer such that $5^\ell = 1 \pmod{2}$. By Theorem 4.7, we can construct the desired good polynomial $g(x)$ over the field $\mathbb{F}_{5^2}$ using (13), since $\ell = 1|2 = s$, and $q = 25 > 20 = n$.

Specifically, the polynomial $g(x)$ partitions the finite field of size 25 into one subset of size $p^{t'} = 5$, and $\frac{p^s - p^{t'}}{mp^{t'}} = \frac{25 - 5}{10} = 2$ subsets of size $mp^{t'} = 10$. To construct a code of length $n = 20$, we therefore use the subset of size $p^{t'} = 5$ to define $g(x)$, and select the two disjoint subsets of size $mp^{t'} = 10$ to form the partition of the evaluation set A of size $n = 20$.

Although the dimension parameter k is not involved in constructing good polynomials and partitions, it is essential for the construction of (n, k, r) LRC codes. By Remark 4.3, for $n = 20$ and $r = 9$, the dimension of the code can take any value $k \leq \frac{nr}{r+1} = \frac{180}{10} = 18$.

4.3.3 Summary of good polynomial constructions

Suppose we aim to construct a good polynomial over a finite field of characteristic p that is constant on disjoint subsets of points of size $r + 1 = mp^{t'}$, where m and p are coprime. To determine which good polynomial construction method is most suitable — whether the annihilator polynomial approach from Section 4.3.1 or the combined construction from Section 4.3.2 — it is necessary to consider the specific values of the underlying parameters m (which governs the multiplicative structure) and t' (which governs the additive structure).

We present a technique that can be used to select an appropriate good polynomial construction based on the value of the m and t' .

Remark 4.8. *With the above setup, three distinct strategies for constructing good polynomials arise, depending on the values of m and t' :*

1. When $t' = 0$ (no additive structure), one can use multiplicative subgroups of some field extension $\mathbb{F}_{p^\ell}$, where ℓ is chosen such that $p^\ell = 1 \pmod{m}$.
2. When $t' > 0$ and $m = 1$ (no multiplicative structure), additive subgroups of a field $\mathbb{F}_{p^s}$ of size $p^{t'}$ are sufficient.
3. When $t', m > 1$ (mixed additive and multiplicative structure), and t' is a multiple of ℓ , where ℓ is the smallest integer such that $p^\ell = 1 \pmod{m}$, the construction is that of Theorem 4.7.

We see that Cases 1 and 2 enable the construction of good polynomials using annihilator polynomials from Section 4.3.1, while Case 3 is the combined construction presented in Section 4.3.2.

The above analysis fixes a prime p and examines which of the two available options should be used to construct a good polynomial $g(x)$ over an extension of $\mathbb{F}_p$. Constructing such a good polynomial $g(x)$ enables the construction of an LRC code via **Construction 1**, since $g(x)$ simultaneously defines a partition of an evaluation set into subsets on which it is constant. Once this partition is obtained, the setup for the construction of an LRC code is essentially complete; it remains only to choose an appropriate dimension k using Remark 4.3).

However, Remark 4.8 does not always yield a good polynomial, and thus it is not always possible to construct an LRC code by combining Remark 4.8 and **Construction 1**. For example, it is impossible to construct a code with locality $r = 5$ over any extension of the field $\mathbb{F}_2$ by applying these two results.

Observe that $r + 1 = 6 = 3 \cdot 2$, so we are in *Case 3* of Remark 4.8, with $m = 3$ and $p' = 2$, i.e., $t' = 1$. The smallest integer ℓ such that $2^\ell = 1 \pmod{3}$ is $\ell = 2$. Since $t' = 1$ is not a multiple of $\ell = 2$, the condition required for *Case 3* fails.

We may attempt alternative factorizations of 6 to determine whether any other cases of Remark 4.8 are applicable:

- *Case 1 attempt*: $6 = 6 \cdot 1$, i.e., $t' = 0, m = 6$. This fails since $\gcd(6, 2) \neq 1$, violating the coprimality condition.
- *Case 2 attempt*: $6 = 1 \cdot 6$, i.e., $t' > 0, m = 1$. This is invalid since 6 is not a power of 2.

Rather than fixing a characteristic p , a natural related question arises:

Given a locality parameter r , for which primes p is it possible to construct a good polynomial — and hence an LRC code (via Construction 1) — over an extensions of $\mathbb{F}_p$?

To address this question, we begin by considering the following two cases:

1. For any locality parameter r such that $r + 1$ is *not* a multiple of a power of a prime p (that is, $r + 1 = mp^0$), one can always construct a good polynomial using *Case 1* of Remark 4.8 over a suitable field extension $\mathbb{F}_{p^\ell} \supseteq \mathbb{F}_p$, where the prime p is coprime to $m = r + 1$, and ℓ is any integer satisfying $p^\ell = 1 \pmod{m}$.
2. For any r such that $r + 1$ is *equal* to a power of a prime p (so $r + 1 = p^{t'}$), one can always construct a good polynomial using *Case 2* of Remark 4.8 over an appropriate field extension of $\mathbb{F}_p$. By “appropriate”, we mean that the extension $\mathbb{F}_{p^s}$ must be large enough to accommodate an evaluation set of size $n \leq p^s$, where n is such that $(r + 1) \mid n$.

Below, we present a table that summarizes the possibilities for constructing good polynomials using Remark 4.8 for various values of the locality parameter r in the range $2 \leq r \leq 20$. In other words, we vary r and examine the possible field sizes q . This allows us to determine, for each locality, which finite fields enable the construction of an LRC code by combining Remark 4.8 and **Construction 1**. A more detailed discussion follows.

	Case 1		Case 2		Case 3	
r	$(t' = 0)$ $r+1=m$	g constructable over $\mathbb{F}_{p^\ell}, p^\ell = 1 \pmod{m}$	$(m = 1)$ $r+1=p^{t'}$	g constructable over $\mathbb{F}_{p^s}$	$r+1=mp^{t'}$	g constructable over $\mathbb{F}_{p^s}$
2	3	$\forall p \neq 3$	3^1	$\mathbb{F}_{3^s}: s \geq 1$	—	—
3	4	$\forall p \neq 2$	2^2	$\mathbb{F}_{2^s}: s \geq 2$	—	—
4	5	$\forall p \neq 5$	5^1	$\mathbb{F}_{5^s}: s \geq 1$	—	—
5	6	$\forall p \neq 2, 3$	—	—	$3 \cdot 2 = 2 \cdot 3$	$\mathbb{F}_{3^s}: s \geq 2$
6	7	$\forall p \neq 7$	7^1	$\mathbb{F}_{7^s}: s \geq 1$	—	—
7	8	$\forall p \neq 2$	2^3	$\mathbb{F}_{2^s}: s \geq 3$	—	—
8	9	$\forall p \neq 3$	3^2	$\mathbb{F}_{3^s}: s \geq 2$	—	—
9	10	$\forall p \neq 2, 5$	—	—	$5 \cdot 2 = 2 \cdot 5$	$\mathbb{F}_{5^s}: s \geq 2$
10	11	$\forall p \neq 11$	11^1	$\mathbb{F}_{11^s}: s \geq 1$	—	—
11	12	$\forall p \neq 2, 3$	—	—	$3 \cdot 2^2 = 4 \cdot 3$	$\mathbb{F}_{2^s}: s \geq 4, 2 \mid s$
12	13	$\forall p \neq 13$	13^1	$\mathbb{F}_{13^s}: s \geq 1$	—	—
13	14	$\forall p \neq 2, 7$	—	—	$7 \cdot 2 = 2 \cdot 7$	$\mathbb{F}_{7^s}: s \geq 2$
14	15	$\forall p \neq 3, 5$	—	—	$5 \cdot 3 = 3 \cdot 5$	—
15	16	$\forall p \neq 2$	2^4	$\mathbb{F}_{2^s}: s \geq 4$	—	—
16	17	$\forall p \neq 17$	17^1	$\mathbb{F}_{17^s}: s \geq 1$	—	—
17	18	$\forall p \neq 2, 3$	—	—	$9 \cdot 2 = 2 \cdot 3^2$	$\mathbb{F}_{3^s}: s \geq 3$
18	19	$\forall p \neq 19$	19^1	$\mathbb{F}_{19^s}: s \geq 1$	—	—
19	20	$\forall p \neq 2, 5$	—	—	$5 \cdot 2^2 = 4 \cdot 5$	$\mathbb{F}_{5^s}: s \geq 2$
20	21	$\forall p \neq 3, 7$	—	—	$7 \cdot 3 = 3 \cdot 7$	$\mathbb{F}_{7^s}: s \geq 2$

Table 1: Possibility of LRC code construction via Remark 4.8 and **Construction 1** for fixed locality

To illustrate the constructions in *Case 3* of the table, we examine four representative values of the locality parameter r . The remaining values can be treated analogously.

Locality $r = 5$.

- Extension of the field $\mathbb{F}_2$: As discussed earlier, it is not possible to construct a code with locality $r = 5$ over any extension of $\mathbb{F}_2$.
- Extension of the field $\mathbb{F}_3$: Here $r + 1 = 6 = 2 \cdot 3$, so $m = 2$, $p' = 3^1$. The smallest integer ℓ such that $3^\ell \bmod 2 = 1$ is $\ell = 1$. Since $t' = 1$ is a multiple of $\ell = 1$, it is possible to construct a code with locality $r = 5$ over an extension of $\mathbb{F}_3$. Note that we require $r + 1 = 6 \mid n$, so we need $3^s \geq 6$, which implies $s \geq 2$.

Locality $r = 9$.

- Extension of the field $\mathbb{F}_2$: Since $r + 1 = 10 = 5 \cdot 2$, we have $m = 5$, $p' = 2^1$. The smallest integer ℓ such that $2^\ell \bmod 5 = 1$ is $\ell = 4$. Since $t' = 1$ is not a multiple of $\ell = 4$, it is not possible to construct a code with locality $r = 9$ over any extension of $\mathbb{F}_2$.
- Extension of the field $\mathbb{F}_5$: Here $r + 1 = 10 = 2 \cdot 5$, so $m = 2$, $p' = 5^1$. The smallest integer ℓ such that $5^\ell \bmod 2 = 1$ is $\ell = 1$, and $t' = 1$ is a multiple of $\ell = 1$. Hence, a code with locality $r = 9$ can be constructed over an extension of $\mathbb{F}_5$. We require $r + 1 = 10 \mid n$, so $5^s \geq 10$, which gives $s \geq 2$.

Locality $r = 11$.

- Extension of the field $\mathbb{F}_2$: Since $r + 1 = 12 = 3 \cdot 2^2$, we have $m = 3$, $p' = 2^2$. The smallest ℓ such that $2^\ell \bmod 3 = 1$ is $\ell = 2$. Since $t' = 2$ is a multiple of $\ell = 2$, it is possible to construct a code with locality $r = 11$ over an extension of $\mathbb{F}_2$. We require $r + 1 = 12 \mid n$, so $2^s \geq 12$, which gives $s \geq 4$. Moreover, in this case $\ell = 2$, and Theorem 4.7 requires $\ell \mid s$, so we must ensure that $2 \mid s$.
- Extension of the field $\mathbb{F}_3$: Here $r + 1 = 12 = 4 \cdot 3$, so $m = 4$, $p' = 3^1$. The smallest ℓ such that $3^\ell \bmod 4 = 1$ is $\ell = 2$, but $t' = 1$ is not a multiple of $\ell = 2$. Hence, it is not possible to construct a code with locality $r = 11$ over any extension of $\mathbb{F}_3$.

Locality $r = 14$.

- Extension of the field $\mathbb{F}_3$: We have $r + 1 = 15 = 5 \cdot 3$, so $m = 5$, $p' = 3^1$. The smallest integer ℓ such that $3^\ell \bmod 5 = 1$ is $\ell = 4$, and since $t' = 1$ is not a multiple of $\ell = 4$, no code with locality $r = 14$ can be constructed over any extension of $\mathbb{F}_3$.
- Extension of the field $\mathbb{F}_5$: Here $r + 1 = 15 = 3 \cdot 5$, so $m = 3$, $p' = 5^1$. The smallest integer ℓ such that $5^\ell \bmod 3 = 1$ is $\ell = 2$, but again $t' = 1$ is not a multiple of $\ell = 2$, so no code with locality $r = 14$ can be constructed over any extension of $\mathbb{F}_5$.

Examining the full table, we highlight three key observations that reflect broader structural patterns.

For locality values $r \in \{2, 3, 4, 6, 7, 8, 10, 12, 15, 16, 18\}$, we observe that the primes over which *Case 1* (which is based on annihilator polynomials for multiplicative subgroups) cannot be used are precisely those that permit constructions under *Case 2* (based on annihilator polynomials for additive subgroups). This correspondence consistently arises when $r + 1$ is either a prime p or a power of a prime p . In such cases, with $r + 1 = p'$, *Case 1* yields good polynomials over extensions of $\mathbb{F}_{\bar{p}}$ for any prime $\bar{p} \neq p$, while *Case 2* applies specifically to extensions of $\mathbb{F}_p$.

Furthermore, we note that for certain values of r , although *Case 1* prohibits the use extensions of a given prime field $\mathbb{F}_p$, *Case 3* still allows the construction of a good polynomial over an extension of the same field. For instance, when $r = 5$, although *Case 1* rules out constructions over extensions of $\mathbb{F}_2$ and $\mathbb{F}_3$, but *Case 3* permits constructions over extensions of $\mathbb{F}_3$. Similar situations occur in the following instances:

- $r = 9$ with $\mathbb{F}_5$,
- $r = 11$ with $\mathbb{F}_2$,
- $r = 13$ with $\mathbb{F}_7$,
- $r = 17$ with $\mathbb{F}_3$,
- $r = 19$ with $\mathbb{F}_5$, and
- $r = 20$ with $\mathbb{F}_7$.

On the other hand, for locality $r = 14$, this phenomenon does not occur: both *Case 1* and *Case 3* prohibit constructions over extensions of $\mathbb{F}_3$ and $\mathbb{F}_5$. This makes $r = 14$ the smallest locality for which neither case yields a good polynomial over these fields.

Although we have seen that Remark 4.8 cannot be applied in certain cases to construct a good polynomial, the proposition below demonstrates that good polynomials do exist in these unresolved settings — provided that one chooses a sufficiently large extension $\mathbb{F}_q$ of $\mathbb{F}_p$.

Proposition 4.9. *Let $\mathbb{F}_q$ be the finite field of size q . There exists a good polynomial of degree $r + 1$ that is constant on at least*

$$\left\lceil \frac{\binom{q}{r+1}}{q^r} \right\rceil$$

subsets of $\mathbb{F}_q$, each of size $r + 1$.

Proof. Consider the set

$$M_{q,r} = \left\{ f = \prod_{i=1}^{r+1} (x - \alpha_i) \in \mathbb{F}_q[x] : \alpha_1, \dots, \alpha_{r+1} \in \mathbb{F}_q \text{ distinct} \right\}.$$

This is the set of all monic degree $r + 1$ polynomials in $\mathbb{F}_q[x]$ with $r + 1$ distinct roots in $\mathbb{F}_q$. The number of such polynomials is $\binom{q}{r+1}$, since there are $\binom{q}{r+1}$ ways to choose $r + 1$ distinct roots from $\mathbb{F}_q$.

A polynomial $f \in M_{q,r}$ can be written as

$$f(x) = x^{r+1} + \sum_{i=0}^r c_i x^i.$$

Define an equivalence relation on $M_{q,r}$: two polynomials are equivalent if they differ by a constant, i.e.,

$$f(x) \sim g(x) \iff f(x) = g(x) + c \quad \text{for some } c \in \mathbb{F}_q.$$

To see this, observe that:

- *Reflexivity:* For $c = 0 \in \mathbb{F}_q$, $f(x) = f(x) + 0$, so $f(x) \sim f(x)$.
- *Symmetry:* If $f(x) \sim g(x)$, then $f(x) = g(x) + c$ for some $c \in \mathbb{F}_q$. Rearranging gives $g(x) = f(x) - c$, and since $-c \in \mathbb{F}_q$, we have $g(x) \sim f(x)$.

- *Transitivity*: Suppose $f(x) \sim g(x)$ and $g(x) \sim h(x)$. That is, $f(x) = g(x) + c_1$ and $g(x) = h(x) + c_2$ for $c_1, c_2 \in \mathbb{F}_q$. Then $f(x) = (h(x) + c_2) + c_1 = h(x) + (c_1 + c_2)$, and since $c_1 + c_2 \in \mathbb{F}_q$, we conclude that $f(x) \sim h(x)$.

This partitions $M_{q,r}$ into equivalence classes. Each equivalence class of a polynomial

$$f(x) = x^{r+1} + \sum_{i=0}^r c_i x^i$$

is uniquely determined by the coefficients $(c_1, \dots, c_r)$ and hence corresponds to a distinct combination of these r coefficients. The total number of the r -tuples $(c_1, \dots, c_r)$ is q^r and the total number equivalence classes is at most q^r , since not all coefficient combinations may correspond to polynomials in $M_{q,r}$ (i.e., some combinations might not yield $r + 1$ distinct roots).

By the pigeonhole principle, at least one equivalence class must contain at least $\left\lceil \frac{\binom{q}{r+1}}{q^r} \right\rceil$ polynomials. In other words, there exists an equivalence class of size at least $\left\lceil \frac{\binom{q}{r+1}}{q^r} \right\rceil$. Let f be a representative of this class. For any other polynomials g in this class, $f(x) = g(x) + c$ and g has $r+1$ distinct roots $\{\alpha_1, \dots, \alpha_r\}$. It follows that f is constant on the set of zeros of any other polynomial g . Each polynomial g in the equivalence class corresponds to a distinct subset $\{\alpha_1, \dots, \alpha_{r+1}\}$ on which f is constant. Therefore, f is a good polynomial that is constant on at least $\left\lceil \frac{\binom{q}{r+1}}{q^r} \right\rceil$ subsets of $\mathbb{F}_q$ of size $r + 1$. $\square$

Example 4.4. We previously observed that for locality $r = 9$, Remark 4.8 cannot be used to construct a good polynomial over an extension of $\mathbb{F}_2$. Consequently, it is not possible to construct a code with $r = 9$ over extensions of $\mathbb{F}_2$ by combining Remark 4.8 and **Construction 1**. However, by Proposition 4.9, we know that a good polynomial $g(x)$ exists for a suitable $\mathbb{F}_q$. For instance, taking $q = 2^{23}$, there exists a good polynomial of degree $r + 1 = 10$ that is constant on at least

$$\left\lceil \frac{\binom{2^{23}}{10}}{2^{207}} \right\rceil \approx [2.31] = 3$$

disjoint subsets of size 10. Therefore, we can apply **Construction 1** using this good polynomial $g(x)$ to construct an optimal LRC code over $\mathbb{F}_{2^{23}}$. For example, we can choose a code of length $n = 3 \cdot 10 = 30$ (since we have at least 3 disjoint sets), locality $r = 9$, and any dimension $k \leq \frac{nr}{r+1} = \frac{30 \cdot 9}{10} = 27$ as guaranteed by Remark 4.3.

We end this section with a final comment. As the field size q increases, the quantity $\left\lceil \frac{\binom{q}{r+1}}{q^r} \right\rceil$ eventually exceeds $\frac{n}{(r+1)}$, which is the desired number of disjoint sets needed in a partition to construct a code of length n and locality r . This, in turn, further guarantees the existence of a good polynomial that is constant on (at least) $\frac{n}{(r+1)}$ disjoint subsets of $\mathbb{F}_q$. In [12], Tamo and Barg suggest that $q > n(r+1)^r$ suffices. In what follows, we derive a tighter bound, namely $q > nr!$.

To ensure that

$$\left\lceil \frac{\binom{q}{r+1}}{q^r} \right\rceil > \frac{n}{(r+1)}$$

it is sufficient to find a bound on q in terms of the known parameters n and $1 < r < k$, such that

$$\frac{\binom{q}{r+1}}{q^r} > \frac{n}{(r+1)}$$

holds.

The binomial coefficient satisfies

$$\binom{q}{r+1} = \frac{q!}{(q-r-1)!(r+1)!} = \frac{q(q-1)(q-2)\cdots(q-r)}{(r+1)!} = \frac{q^{r+1}(1-\frac{1}{q})(1-\frac{2}{q})\cdots(1-\frac{r}{q})}{(r+1)!}.$$

For sufficiently large $q \gg r$, the terms $\frac{t}{q}$, where $1 \leq t \leq r$, become negligible, so we approximate

$$\binom{q}{r+1} \approx \frac{q^{r+1}}{(r+1)!}.$$

Then,

$$\frac{\binom{q}{r+1}}{q^r} \approx \frac{\frac{q^{r+1}}{(r+1)!}}{q^r} = \frac{q}{(r+1)!}.$$

To ensure

$$\frac{q}{(r+1)!} > \frac{n}{r+1},$$

we solve for q to obtain

$$q > \frac{n(r+1)!}{r+1} = nr!.$$

This yields a tighter bound than $q > n(r+1)^r$, since for any $r \geq 1$ we have

$$(r+1)^r = \underbrace{(r+1) \cdots (r+1)}_{r \text{ times}} > \underbrace{r(r-1) \cdots 1}_{r \text{ terms}} = r!.$$

The above demonstrates that for any given locality r , even if a good polynomial cannot be constructed using Remark 4.8 over an extension of a prime field $\mathbb{F}_p$, there always exists a sufficiently large finite field $\mathbb{F}_q \supset \mathbb{F}_p$ over which such a good polynomial exists. Consequently, an LRC code can be constructed via **Construction 1** over a sufficiently large finite field $\mathbb{F}_q$.

Example 4.5. Suppose we wish to construct a $(30, 20, 9)$ LRC code over an extension of $\mathbb{F}_2$. If we take $q = 2^{35} \approx (3.44) \cdot 10^{10} > 3 \cdot 10^{10} = n(r+1)^r$, then

$$\left\lceil \frac{\binom{2^{35}}{10}}{2^{315}} \right\rceil = 9469.$$

If we choose $q = 2^{24} \approx (1.68) \cdot 10^7 > (1.09) \cdot 10^7 \approx 30 \cdot 9! = nr!$, then

$$\left\lceil \frac{\binom{2^{24}}{10}}{2^{216}} \right\rceil = 5.$$

Therefore, with either of these choices of q , we exceed the required number $\frac{n}{r+1} = \frac{30}{10} = 3$ of disjoint sets needed for the construction.

4.4 Closing remarks on Construction 1

In this section, we conclude our discussion of **Construction 1**. Remark 4.10 explores the connection between **Construction 1** and Reed-Solomon codes, Remark 4.11 presents a reformulation of the encoding polynomial $f_a(x)$, and Remark 4.12 removes the original divisibility condition $r \mid k$.

Remark 4.10. *Construction 1 extends classical Reed-Solomon (RS) codes in that it also encodes messages by evaluating polynomials defined by the message vector.*

When the locality parameter r equals the dimension k , the construction in fact reduces to standard RS codes. To see this, observe that when $r = k$, each coefficient polynomial

$$f_i(x) = \sum_{j=0}^{\frac{k}{r}-1} a_{i,j} g(x)^j$$

as defined in (5), becomes the constant term $a_{i,0} =: a_i$, and the encoding polynomial simplifies to

$$f_{\mathbf{a}}(x) = \sum_{i=0}^{k-1} a_i x^i$$

which coincides with the classical RS encoding polynomial seen in Section 2.3.

Therefore, when $r = k$, the **Construction 1** reproduces the RS code construction and does not rely on the existence of a good polynomial. Consequently, for RS codes, the evaluation set A can be any arbitrary subset of $\mathbb{F}_q$, with no need to impose a partitioning structure.

In contrast, when $r < k$, the locality constraint enforces a specific structure on the choice of evaluation points, and the existence of a suitable good polynomial $g(x)$ becomes essential.

Remark 4.11. The encoding polynomial $f_{\mathbf{a}}$ in (6) can also be expressed using a single summation over a combined linear indexing of the message vector $\mathbf{a} = (a_0, \dots, a_{k-1}) \in \mathbb{F}_q^k$. Specifically, it can be written as

$$f_{\mathbf{a}}(x) = \sum_{\substack{m=0 \\ m \neq r \bmod (r+1)}}^{k-\frac{k}{r}-2} a_m g(x)^{\lfloor \frac{m}{r+1} \rfloor} x^{m \bmod (r+1)}. \quad (14)$$

This alternative expression arises from flattening the original two-dimensional indexing $a_{i,j}$ (used in the double sum formulations of $f_{\mathbf{a}}$) into a single index $m = i + j(r+1)$ for $i = 0, \dots, r-1$ and $j = 0, \dots, \frac{k}{r} - 1$. Since $i < r$, this indexing scheme excludes values $m = r \bmod (r+1)$.

The new formulation of $f_{\mathbf{a}}(x)$ in (14) is equivalent to the original one in (6). To see this, recall that in the original construction, we have the message vector $\mathbf{a} \in \mathbb{F}_q^k$ consisting of entries $a_{i,j}$, where $i = 0, \dots, r-1$ and $j = 0, \dots, \frac{k}{r} - 1$, and the corresponding encoding polynomial is defined as

$$f_{\mathbf{a}}(x) = \sum_{i=0}^{r-1} \sum_{j=0}^{\frac{k}{r}-1} a_{i,j} g(x)^j x^i.$$

To re-express this as a single sum, we define a new index $m = i + j(r+1)$, and relabel each coefficient $a_{i,j}$ as a_m . This mapping is injective, since different pairs (i, j) yield different values of m due to $i < r$. Using this reindexing, we recover the original i and j from m by computing

$$j = \left\lfloor \frac{m}{r+1} \right\rfloor, \quad i = m \bmod (r+1).$$

Therefore, the monomial $g(x)^j x^i$ becomes

$$g(x)^{\lfloor \frac{m}{r+1} \rfloor} x^{m \bmod (r+1)},$$

and the encoding polynomial can be rewritten as

$$f_a(x) = \sum_{\substack{m=0 \\ m \neq r \bmod (r+1)}}^{k + \frac{k}{r} - 2} a_m g(x) \lfloor \frac{m}{r+1} \rfloor x^{m \bmod (r+1)}.$$

The restriction $m \neq r \bmod (r+1)$ arises naturally: for any valid pair (i, j) in the original double-indexed form, we have $i \in \{0, \dots, r-1\}$, and so $m \bmod (r+1) = i < r$. Hence, values of $m = r \bmod (r+1)$ do not correspond to any valid message coefficient $a_{i,j}$, and are excluded from the sum. This ensures that both formulations produce the same polynomial, simply written in different indexing conventions.

To relate the indexing $(a_0, \dots, \alpha_{k-1})$ to the structure of the summation, we make the following observation. From the proof of Theorem 4.2, we have

$$k + \frac{k}{r} - 2 = \left(\frac{k}{r} - 1 \right) (r+1) + (r-1).$$

Dividing both sides by $r+1$, we obtain

$$\frac{k + \frac{k}{r} - 2}{r+1} = \left(\frac{k}{r} - 1 \right) + \frac{r-1}{r+1}.$$

Since $\frac{r-1}{r+1} < 1$, it follow that

$$\left\lfloor \frac{k + \frac{k}{r} - 2}{r+1} \right\rfloor = \frac{k}{r} - 1.$$

Claim. In the set $\{0, 1, \dots, k + \frac{k}{r} - 2\}$ there are $\frac{k}{r} - 1$ numbers corresponding to $r \bmod (r+1)$.

Proof. The numbers in the set $\{0, 1, \dots, k + \frac{k}{r} - 2\}$ that satisfy $m = r \bmod (r+1)$ are of the form

$$m = (r+1)t + r$$

where $t \in \mathbb{Z}_{\geq 0}$, and

$$(r+1)t + r \leq k + \frac{k}{r} - 2.$$

Solving for t we obtain

$$t \leq \frac{k + \frac{k}{r} - 2 - r}{r+1} = \frac{rk + k - 2r - r^2}{r(r+1)} < \frac{k(r+1) - r(r+1)}{r(r+1)} = \frac{k-r}{r} = \frac{k}{r} - 1,$$

Thus $t = 0, 1, \dots, \frac{k}{r} - 2$, which is exactly $\frac{k}{r} - 1$ numbers, as required. $\square$

Hence, the number of values of $m \in \{0, \dots, k + \frac{k}{r} - 2\}$ that must be excluded is exactly $\frac{k}{r} - 1$. Once these are removed, we are left with precisely k valid indices for a_m , matching the total number of message symbols $a_{i,j}$ in the original formulation.

Since the reformulation in (14) is equivalent to the original encoding polynomial in **Construction 1**, it can be used interchangeably to construct optimal LRC codes via **Construction 1**. When this alternative form of the encoding polynomial is used, the local recovery mechanism in Section 4.2.3 operates in exactly the same way, simply applied to this new representation.

We present an example that employs the reformulated encoding polynomial $f_a(x)$.

Example 4.6. We use **Construction 1** to construct an optimal $(16, 9, 3)$ LRC code with minimum distance $d = 6$ over $\mathbb{F}_{17}$.

Note that 4 is an $(r + 1) = 4^{\text{th}}$ root of unity modulo 17, and thus generates a multiplicative subgroup of order 4:

$$H = \{1, 4, 16, 13\} \subset \mathbb{F}_{17}^\times.$$

The cosets of H are:

$$\begin{aligned} A_1 &= H = \{1, 4, 16, 13\}, \\ A_2 &= 2H = \{2, 8, 15, 9\}, \\ A_3 &= 3H = \{3, 12, 14, 5\}, \\ A_4 &= 6H = \{6, 7, 11, 10\}. \end{aligned}$$

yielding a partition of $\mathbb{F}_{17}^\times$ into:

$$\mathcal{A} = \{A_1, A_2, A_3, A_4\}.$$

The annihilator polynomial of H is $x^4 - 1$, so we pick the good polynomial $g(x) = x^4$, which is also constant on each $A_i \in \mathcal{A}$ and defines an equivalent code by Remark 4.5.

Let the message vector be $\mathbf{a} = (a_0, a_1, a_2, a_4, a_5, a_6, a_8, a_9, a_{10}) \in \mathbb{F}_{17}^9$. The encoding polynomial (as in Equation (14)) is given by:

$$\begin{aligned} f_{\mathbf{a}}(x) &= \sum_{\substack{m=0 \\ m \not\equiv 3 \pmod{4}}}^{10} a_m g(x)^{\lfloor \frac{m}{4} \rfloor} x^{m \bmod 4} \\ &= a_0 g(x)^0 x^0 + a_1 g(x)^0 x^1 + a_2 g(x)^0 x^2 + a_4 g(x)^1 x^0 + a_5 g(x)^1 x^1 \\ &\quad + a_6 g(x)^1 x^2 + a_8 g(x)^2 x^0 + a_9 g(x)^2 x^1 + a_{10} g(x)^2 x^2 \\ &= a_0 + a_1 x + a_2 x^2 + a_4 x^4 + a_5 x^5 + a_6 x^6 + a_8 x^8 + a_9 x^9 + a_{10} x^{10}. \end{aligned}$$

We observe that the encoding polynomial can simply be written as:

$$f_{\mathbf{a}}(x) = \sum_{\substack{m=0 \\ m \not\equiv 3,7}}^{10} a_m x^m. \quad (15)$$

This expression can be equivalently regrouped as

$$f_{\mathbf{a}}(x) = f_0(x) + f_1(x)x + f_2(x)x^2$$

where the coefficient polynomials $f_i(x)$ are defined as:

$$f_i(x) = \sum_{j=0}^{\frac{k}{r}-1} a_{i+j(r+1)} g(x)^j \quad \text{for } i = 0, 1, 2.$$

That is,

$$f_0(x) = a_0 + a_4 x^4 + a_8 x^8, \quad f_1(x) = a_1 + a_5 x^4 + a_9 x^8, \quad f_2(x) = a_2 + a_6 x^4 + a_{10} x^8.$$

Note that in (15) we exclude the monomials x^3 and x^7 with coefficients a_3 and a_7 (i.e., $m = 3 \pmod{4}$) in the encoding polynomial. Consequently, the coefficients a_3 and a_7 are omitted from the message vector $\mathbf{a}$. This restriction simplifies notation and ensures that the resulting encoding polynomial aligns with the required structure.

The final codeword is obtained by evaluating $f_{\mathbf{a}}(x)$ at all $x \in \mathbb{F}_{17}^\times$.

Remark 4.12. In the original formulation of **Construction 1**, it was assumed that the locality parameter r divides the dimension k . However, this condition can be removed without altering the core structure of the construction. Specifically, when $r \nmid k$, we redefine the coefficient polynomials $f_i(x)$ from (5) as follows:

$$f_i(x) = \sum_{j=0}^{s(k,r,i)} a_{i,j} g(x)^j, \quad i = 0, 1, \dots, r-1, \quad (16)$$

where the function $s(k, r, i)$ is given by:

$$s(k, r, i) = \begin{cases} \left\lfloor \frac{k}{r} \right\rfloor & \text{if } i < k \bmod r, \\ \left\lfloor \frac{k}{r} \right\rfloor - 1 & \text{if } i \geq k \bmod r, \end{cases}$$

and $k \bmod r \in \{0, 1, \dots, r-1\}$ denotes the integer remainder of k upon division by r (this notation will be used throughout the rest of the remark).

Our goal is to demonstrate that this modified formulation remains compatible with the application of **Construction 1**. The derivations below were obtained independently, although similar reasoning appears in [6].

In the original construction, the encoding polynomial $f_{\mathbf{a}}$ is expressed in terms of k message symbols, which are distributed across r coefficient polynomials. We verify that, under the revised definition in (16), the total number of coefficients in $f_{\mathbf{a}}$ remains exactly k , even when $r \nmid k$.

We distinguish between the following two index ranges:

- For indices $i < k \bmod r$, each polynomial $f_i(x)$ has $\left\lfloor \frac{k}{r} \right\rfloor + 1$ coefficients.
- For indices $i \geq k \bmod r$, each polynomial $f_i(x)$ has $\left\lfloor \frac{k}{r} \right\rfloor$ coefficients.

Since there are $k \bmod r$ polynomials of the first kind and $r - (k \bmod r)$ of the second kind, the total number of coefficients in $f_{\mathbf{a}}$ is:

$$\begin{aligned} (k \bmod r) \left(\left\lfloor \frac{k}{r} \right\rfloor + 1 \right) + (r - (k \bmod r)) \left\lfloor \frac{k}{r} \right\rfloor &= (k \bmod r) \left\lfloor \frac{k}{r} \right\rfloor + (k \bmod r) + r \left\lfloor \frac{k}{r} \right\rfloor - (k \bmod r) \left\lfloor \frac{k}{r} \right\rfloor \\ &= (k \bmod r) + r \left\lfloor \frac{k}{r} \right\rfloor \\ &= k. \end{aligned}$$

This confirms that the r coefficient polynomials defined in (16) are completely determined by k message symbols.

We now analyze the degree of the resulting encoding polynomial $f_{\mathbf{a}}$. Observe that

$$\begin{aligned} \deg(f_{\mathbf{a}}) &\leq \max_{0 \leq i < r} (\deg(f_i) + i) = \left\lfloor \frac{k}{r} \right\rfloor (r+1) + (k \bmod r) - 1 \\ &= r \left\lfloor \frac{k}{r} \right\rfloor + \left\lfloor \frac{k}{r} \right\rfloor + \left(k - r \left\lfloor \frac{k}{r} \right\rfloor \right) - 1 \\ &= \left\lfloor \frac{k}{r} \right\rfloor - 1 + k - 1 \\ &= k + \left\lfloor \frac{k}{r} \right\rfloor - 2. \end{aligned}$$

where in the second equality we use the identity $k \bmod r = k - r \left\lfloor \frac{k}{r} \right\rfloor$, and in the third equality, we apply the fact that $\left\lfloor \frac{k}{r} \right\rfloor = \left\lceil \frac{k}{r} \right\rceil - 1$ when $r \nmid k$.

Following the same reasoning as in the proof of Theorem 4.2, we obtain

$$k + \left\lceil \frac{k}{r} \right\rceil - 2 \leq \frac{nr}{r+1} + \left\lceil \frac{n}{r+1} \right\rceil - 2 = \frac{nr}{r+1} + \frac{n}{r+1} - 2 = n - 2$$

which implies that

$$\deg(f_{\mathbf{a}}) \leq n - 2.$$

Thus, the code has dimension k , and the minimum distance satisfies:

$$d(C) \geq n - \max_{f_{\mathbf{a}}, \mathbf{a} \in \mathbb{F}_q^k} \deg(f_{\mathbf{a}}) = n - k - \left\lceil \frac{k}{r} \right\rceil + 2.$$

which, together with the Singleton-type bound (4), show that equality is achieved.

The local recovery procedure described in Section 4.2.3 applies analogously in this setting: it operates in precisely the same manner, now applied to the adjusted polynomial representation.

Hence, **Construction 1** remains valid even when $r \nmid k$.

We provide an example illustrating the case where r does not divide k .

Example 4.7. We construct an optimal $(10, 5, 4)$ LRC code with minimum distance $d = 5$ over $\mathbb{F}_{11}$.

For each $i = 0, 1, 2, 3$, we compute the values of the function $s(k, r, i)$ using $k = 5$, $r = 4$. Note that $5 = 4 + 1$ (i.e., $5 \bmod 4 = 1$):

$$\begin{aligned} s(5, 4, 0) &= \left\lfloor \frac{5}{4} \right\rfloor = 1 \quad \text{since } 0 < 1, \\ s(5, 4, 1) &= \left\lfloor \frac{5}{4} \right\rfloor - 1 = 0 \quad \text{since } 1 = 1 \\ s(5, 4, 2) &= \left\lfloor \frac{5}{4} \right\rfloor - 1 = 0 \quad \text{since } 2 > 1, \\ s(5, 4, 3) &= \left\lfloor \frac{5}{4} \right\rfloor - 1 = 0 \quad \text{since } 3 > 1. \end{aligned}$$

For each i , we define the coefficient polynomial as:

$$f_0(x) = a_{0,0} + a_{0,1}g(x), \quad f_1(x) = a_{1,0}, \quad f_2(x) = a_{2,0}, \quad f_3(x) = a_{3,0}.$$

This yields the encoding polynomial:

$$f_{\mathbf{a}}(x) = (a_{0,0} + a_{0,1}g(x)) + a_{1,0}x + a_{2,0}x^2 + a_{3,0}x^3.$$

We take the partition $\mathcal{A} = \{A_1, A_2\}$, where

$$\begin{aligned} A_1 &= \{1, 4, 5, 9, 3\}, \\ A_2 &= \{2, 8, 10, 7, 6\}, \end{aligned}$$

with A_1 being the multiplicative subgroup of $\mathbb{F}_{11}^\times$ of order 5, generated by 4 (that is, an $(r+1) = 5^{\text{th}}$ root of unity modulo 11) and A_2 its coset $2A_1$.

The appropriate good polynomial in this case is $g(x) = x^5$, derived from the annihilator polynomial $x^5 - 1$. Therefore, the encoding polynomial becomes:

$$f_a(x) = a_{0,0} + a_{1,0}x + a_{2,0}x^2 + a_{3,0}x^3 + a_{0,1}x^5.$$

5 Construction 2 — a generalization of Construction 1

We present a generalization of **Construction 1**, as introduced in Section III.C of [12]. To this end, we study the mapping from the set of polynomials of the form (6) to $\mathbb{F}_q^n$. From this point onward, we adopt the notation $\mathbb{F}$ to denote the finite field $\mathbb{F}_q$.

5.1 A commutative $\mathbb{F}$ -algebra

Consider a subset $A \subseteq \mathbb{F}$ partitioned into m disjoint subsets A_i , and let $\mathcal{A}$ denote this partition. We define a special set of polynomials, denoted $\mathbb{F}_{\mathcal{A}}[x]$, consisting of polynomials over $\mathbb{F}$ of degree less than $|A|$ that are constant on each subset of the partition $\mathcal{A}$. That is,

$$\mathbb{F}_{\mathcal{A}}[x] = \left\{ f \in \mathbb{F}[x] : f \text{ is constant on each } A_i \text{ for } i = 1, \dots, m, \text{ and } \deg f < |A| \right\}. \quad (17)$$

We also define the *annihilator* of A . Recall the annihilator polynomial (12) defined in Section 4.3.1. A similar idea applies here. The *annihilator* of A is the monic polynomial $h(x)$ of minimal degree such that $h(a) = 0$ for all $a \in A$, i.e.,

$$h(x) = \prod_{a \in A} (x - a)$$

Equipped with standard polynomial addition and multiplication modulo $h(x)$, the set $\mathbb{F}_{\mathcal{A}}[x]$ forms a *finite-dimensional commutative $\mathbb{F}$ -algebra* with identity. Here, the notation $f \bmod h$ is used to denote the canonical representative of the equivalence class $[f]$. In other words, $f \bmod h$ is the remainder of f upon division by h .

Definition 5.1 (Commutative R -algebra). *Let R be a commutative ring. An R -algebra S is a set that is both a ring and an R -module, where:*

1. *Ring addition and module addition coincide (i.e., they are the same operation),*
2. *Scalar multiplication satisfies the compatibility condition:*

$$r(xy) = (rx)y = x(ry) \quad \text{for all } r \in R \text{ and } x, y \in S.$$

If S is a commutative ring, then S is called a commutative R -algebra.

When R is a field $\mathbb{F}$, the above definition simplifies to S being both a ring and a vector space over $\mathbb{F}$, with identical additive structures. The compatibility condition remains:

$$\lambda(xy) = (\lambda x)y = x(\lambda y) \quad \text{for all } \lambda \in \mathbb{F} \text{ and } x, y \in S.$$

Again, if S is a commutative ring, then S is a commutative $\mathbb{F}$ -algebra.

An $\mathbb{F}$ -algebra S is *finite-dimensional* if it is finite-dimensional as a vector space over $\mathbb{F}$. The dimension of the $\mathbb{F}$ -algebra S is then the dimension of the vector space S over $\mathbb{F}$.

We verify that $\mathbb{F}_{\mathcal{A}}[x]$ is a finite-dimensional commutative $\mathbb{F}$ -algebra with identity. Note that $\mathbb{F}_{\mathcal{A}}[x]$ is a subset $\mathbb{F}[x]$, the ring of polynomials over $\mathbb{F}$, and $\mathbb{F}[x]$ is a vector space over $\mathbb{F}$.

1. $\mathbb{F}_{\mathcal{A}}[x]$ is a finite-dimensional vector space over $\mathbb{F}$

- *Closure under addition:* If $f, g \in \mathbb{F}_{\mathcal{A}}[x]$, then $f+g$ is constant on each A_i (since both f and g are). Moreover, $\deg(f+g) \leq \max(\deg(f), \deg(g)) < |A|$, so $f+g \in \mathbb{F}_{\mathcal{A}}[x]$.
- *Closure under scalar multiplication:* For $\lambda \in \mathbb{F}$ and $f \in \mathbb{F}_{\mathcal{A}}[x]$, λf is constant on each A_i , and $\deg(\lambda f) = \deg(f) < |A|$. Hence, $\lambda f \in \mathbb{F}_{\mathcal{A}}[x]$.
- *Zero Vector:* The zero polynomial is constant on each A_i and has degree less than $|A|$, so it belongs to $\mathbb{F}_{\mathcal{A}}[x]$.
- *Finite dimension:* Since $\mathbb{F}_{\mathcal{A}}[x]$ is a subspace of $\mathbb{F}[x]_{<|A|}$ (the space of polynomials of degree $< |A|$), it has dimension at most $|A|$. We will later see that the additional condition requiring every $f \in \mathbb{F}_{\mathcal{A}}[x]$ to be constant on each of the m subsets $A_i \in \mathcal{A}$ implies that the dimension is $\dim(\mathbb{F}_{\mathcal{A}}[x]) = m$.

2. $\mathbb{F}_{\mathcal{A}}[x]$ is a commutative ring with identity

- *Addition:* As noted above, $\mathbb{F}_{\mathcal{A}}[x]$ is closed under polynomial addition, which is associative and commutative. The zero polynomial acts as the additive identity, and every $f \in \mathbb{F}_{\mathcal{A}}[x]$ has an additive inverse $-f \in \mathbb{F}_{\mathcal{A}}[x]$.
- *Multiplication modulo $h(x)$:*
 - For $f, g \in \mathbb{F}_{\mathcal{A}}[x]$, the polynomial product fg is constant on each A_i (since both f and g are). Reducing modulo $h(x)$ preserves this property because $h(a) = 0$ for all $a \in A$. Thus, $fg \bmod h$ remains constant on each A_i . Furthermore, by definition, $\deg(fg \bmod h) < |A|$, hence $fg \bmod h \in \mathbb{F}_{\mathcal{A}}[x]$.
 - Multiplication modulo $h(x)$ is associative, commutative, and distributive over addition (as inherited from $\mathbb{F}[x]$).
 - The constant polynomial $1 \in \mathbb{F}_{\mathcal{A}}[x]$ serves as a multiplicative identity for all $f \in \mathbb{F}_{\mathcal{A}}[x]$.

3. Compatibility of scalar multiplication with ring multiplication

Usual polynomial multiplication is $\mathbb{F}$ -bilinear, so for any $f, g \in \mathbb{F}[x]$ and $\lambda \in \mathbb{F}$, we have:

$$\lambda(fg) = (\lambda f)g = f(\lambda g).$$

Since reduction modulo $h(x)$ is a ring homomorphism, this equality is preserved in $\mathbb{F}_{\mathcal{A}}[x]$. That is,

$$\lambda(fg \bmod h) = (\lambda(f \bmod h))(g \bmod h) = (f \bmod h)(\lambda(g \bmod h)).$$

Since every polynomial in $\mathbb{F}_{\mathcal{A}}[x]$ is constant on each subset A_i of the partition $\mathcal{A}$, it is convenient to denote the value taken by a polynomial $f \in \mathbb{F}_{\mathcal{A}}[x]$ on a block $A_i \in \mathcal{A}$ by $f(A_i)$. Additionally, from this point on, we will use the notation fg to denote $fg \bmod h$.

The following proposition describes several structural properties of the $\mathbb{F}$ -algebra $\mathbb{F}_{\mathcal{A}}[x]$.

Proposition 5.2. 1. For a nonconstant polynomial $f \in \mathbb{F}_{\mathcal{A}}[x]$, which, by construction, is constant on each subset $A_i \in \mathcal{A}$, the degree of f is bounded below by the size of the largest subset in the partition. Specifically,

$$\max_i |A_i| \leq \deg(f) < |A|.$$

2. There exist m polynomials $f_1, \dots, f_m \in \mathbb{F}_{\mathcal{A}}[x]$ satisfying

$$f_i(A_j) = \delta_{i,j}, \quad \deg(f_i) < |A| \quad \text{for } i, j = 1, \dots, m,$$

where $\delta_{i,j}$ is the Kronecker delta. These polynomials are explicitly given by

$$f_i(x) = \sum_{a \in A_i} \prod_{b \in A \setminus \{a\}} \frac{x-b}{a-b}. \quad (18)$$

The set $\{f_1, \dots, f_m\}$ forms a basis for $\mathbb{F}_{\mathcal{A}}[x]$, and the dimension of $\mathbb{F}_{\mathcal{A}}[x]$ is $\dim(\mathbb{F}_{\mathcal{A}}[x]) = m$.

3. Given m distinct nonzero scalars $\alpha_1, \dots, \alpha_m \in \mathbb{F}$, the polynomial $g \in \mathbb{F}_{\mathcal{A}}[x]$ of degree less than $|A|$ satisfying $g(A_i) = \alpha_i$ for each $i = 1, \dots, m$ is given explicitly by

$$g(x) = \sum_{i=1}^m \alpha_i \sum_{a \in A_i} \prod_{b \in A \setminus \{a\}} \frac{x-b}{a-b},$$

i.e., a linear combination of the basis polynomials f_i , scaled by the values α_i . The set $\{1, g, g^2, \dots, g^{m-1}\}$ forms another basis of $\mathbb{F}_{\mathcal{A}}[x]$.

4. There exist m distinct integers $0 = d_0 < d_1 < \dots < d_{m-1} < |A|$ such that every polynomial in $\mathbb{F}_{\mathcal{A}}[x]$ has degree d_i for some i . In other words, these degrees d_i correspond to the degrees of the elements in a basis of $\mathbb{F}_{\mathcal{A}}[x]$.

Proof. 1. Consider $f \in \mathbb{F}_{\mathcal{A}}[x]$, which is constant on each subset $A_i \in \mathcal{A}$, and nonconstant overall. The polynomial $\tilde{f}(x) := f(x) - f(A_i)$ vanishes for all $x \in A_i$. Therefore, $\tilde{f}$ has at least $|A_i|$ distinct zeros in $\mathbb{F}$, implying that $\deg(\tilde{f}) = \deg(f) \geq |A_i|$. Since this argument must apply to every $A_i \in \mathcal{A}$, taking the maximum over all i yields the result.

2. We first explain why the m polynomials $f_1, \dots, f_m$ defined in (18) belong to $\mathbb{F}_{\mathcal{A}}[x]$. For a given $i \in \{1, \dots, m\}$, each term in the sum defining $f_i(x)$ is a Lagrange basis polynomial. For a fixed $a \in A_i$ in the sum of (18), the term

$$\ell_a(x) := \prod_{b \in A \setminus \{a\}} \frac{x-b}{a-b} \quad (19)$$

satisfies $\ell_a(a) = 1$ and $\ell_a(x) = 0$ for all $x \in A \setminus \{a\}$. Choosing $x = \tilde{a} \in A_i$, we have

$$f_i(\tilde{a}) = \sum_{a \in A_i} \ell_a(\tilde{a}),$$

where when $a = \tilde{a}$, the corresponding term in the sum is 1, and otherwise the terms are 0. Thus, $f_i(\tilde{a}) = 1$ for all $\tilde{a} \in A_i$, and hence $f_i(A_i) = 1$. In contrast, if $x = \bar{a} \in A_j$, for $j \neq i$, then for each $a \in A_i$ in the sum of (18),

$$\ell_a(\bar{a}) = 0$$

and so $f_i(A_j) = 0$. Therefore, the polynomials f_i defined in (18) are constant on each subset $A_i \in \mathcal{A}$, taking values 0 or 1.

Clearly, the degree of each f_i is strictly less than $|A|$, since each Lagrange basis polynomial (19) is a product of $|A| - 1$ linear factors, hence is of degree less than $|A|$. It follows that $f_i \in \mathbb{F}_{\mathcal{A}}[x]$.

To show that the polynomials $f_1, \dots, f_m$ form a basis for $\mathbb{F}_{\mathcal{A}}[x]$, suppose

$$\sum_{i=1}^m c_i f_i(x) = 0$$

for some scalars $c_i \in \mathbb{F}$. Then, for any $j = 1, \dots, m$,

$$\sum_{i=1}^m c_i f_i(A_j) = \sum_{i=1}^m c_i \delta_{i,j} = c_j = 0.$$

Hence, $c_1 = \dots = c_m = 0$, and the set $\{f_1, \dots, f_m\}$ is linearly independent.

By definition, the polynomials $f_1, \dots, f_m$ span $\mathbb{F}_{\mathcal{A}}[x]$. Indeed, for any $f \in \mathbb{F}_{\mathcal{A}}[x]$, the polynomial is constant on each A_i , taking the value $f(A_i)$. It follows that

$$f(x) = \sum_{i=1}^m f(A_i) f_i(x).$$

We conclude that the set $\{f_1, \dots, f_m\}$ is a basis of $\mathbb{F}_{\mathcal{A}}[x]$. Consequently, $\dim(\mathbb{F}_{\mathcal{A}}[x]) = m$. We emphasize that the dimension of $\mathbb{F}_{\mathcal{A}}[x]$ is equal to the number of sets in the partition $\mathcal{A}$.

3. By definition, the polynomial $g \in \mathbb{F}_{\mathcal{A}}[x]$, and since $\mathbb{F}_{\mathcal{A}}[x]$ is closed under multiplication modulo the annihilator $h(x)$, it follows that $g^{j-1} \in \mathbb{F}_{\mathcal{A}}[x]$ for $j = 1, \dots, m$.

From part (2), we know that $\dim(\mathbb{F}_{\mathcal{A}}[x]) = m$. Therefore, to establish that $\{1, g, g^2, \dots, g^{m-1}\}$ forms a basis of $\mathbb{F}_{\mathcal{A}}[x]$, it suffices to show that these m polynomials are linearly independent.

Suppose there exist scalars $\beta_1, \dots, \beta_m \in \mathbb{F}$ such that

$$\sum_{j=1}^m \beta_j g^{j-1}(x) = 0.$$

Evaluating this identity on each subset $A_i \in \mathcal{A}$, we obtain:

$$\sum_{j=1}^m \beta_j g^{j-1}(A_i) = 0 \quad \text{for all } i = 1, \dots, m.$$

This yields a linear system in the coefficient vector $(\beta_1, \dots, \beta_m)^T$, expressed as

$$V \cdot (\beta_1, \dots, \beta_m)^T = \mathbf{0},$$

where V is the $m \times m$ matrix defined by

$$V = (g^{j-1}(A_i))_{1 \leq i, j \leq m},$$

which is a Vandermonde matrix constructed from the values $g(A_1), \dots, g(A_m)$. Explicitly, V has the form:

$$V = \begin{bmatrix} 1 & g(A_1) & g(A_1)^2 & \dots & g(A_1)^{m-1} \\ 1 & g(A_2) & g(A_2)^2 & \dots & g(A_2)^{m-1} \\ 1 & g(A_3) & g(A_3)^2 & \dots & g(A_3)^{m-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & g(A_m) & g(A_m)^2 & \dots & g(A_m)^{m-1} \end{bmatrix}.$$

Since by construction the values $g(A_1), \dots, g(A_m)$ are all distinct, nonzero elements of $\mathbb{F}$, the matrix V is invertible. Hence, the only solution to the system is $\beta_1 = \dots = \beta_m = 0$, proving that $\{1, g, g^2, \dots, g^{m-1}\}$ is linearly independent.

4. Let $\{f_0, \dots, f_{m-1}\}$ be any basis of $\mathbb{F}_{\mathcal{A}}[x]$. These polynomials may initially have non-distinct de-

degrees. In that case, we can construct a new basis $\{\tilde{f}_0, \dots, \tilde{f}_{m-1}\}$ for $\mathbb{F}_{\mathcal{A}}[x]$ by applying linear operations on the f_i 's, such that the degrees $\deg(\tilde{f}_i)$ are all distinct, and $d_0 := \deg(\tilde{f}_0) = 0$.

Specifically, each polynomial f_i can be expressed as a linear combination of monomials of degree less than $|A|$, i.e.,

$$f_i(x) = \sum_{j=0}^{|A|-1} c_{i,j} x^j,$$

for some coefficients $c_{i,j} \in \mathbb{F}$. We represent f_i by its coefficient vector $\mathbf{c}_i = (c_{i,0}, \dots, c_{i,|A|-1}) \in \mathbb{F}^{|A|}$.

Let M be the $m \times |A|$ matrix whose i^{th} row is the vector $\mathbf{c}_i$. Applying Gaussian elimination to M , we obtain a matrix $\tilde{M}$ in reduced row-echelon form. The rows of $\tilde{M}$ correspond to new vectors $\tilde{\mathbf{c}}_0, \dots, \tilde{\mathbf{c}}_{m-1}$, which represent new polynomials $\tilde{f}_0, \dots, \tilde{f}_{m-1}$. These new polynomials are linear combinations of the original basis elements f_i . Since row operations preserve linear independence and span, the polynomials $\tilde{f}_0, \dots, \tilde{f}_{m-1}$ still form a basis of $\mathbb{F}_{\mathcal{A}}[x]$. Moreover, the leading entries of the rows in $\tilde{M}$ occur in distinct columns. Since these columns correspond to distinct monomial degrees, this ensures $\deg(\tilde{f}_i) \neq \deg(\tilde{f}_j)$ for $i \neq j$; that is, the degrees of $\tilde{f}_i$ s are distinct.

Reordering the new basis polynomials $\{\tilde{f}_i\}_{i=0}^{m-1}$ by increasing degree, we write $d_i = \deg(\tilde{f}_i)$ with $d_0 < d_1 < \dots < d_{m-1} < |A|$. Since $\mathbb{F}_{\mathcal{A}}[x]$ contains the constant polynomials (which are constant on every A_i), it must hold that $d_0 = 0$.

Thus, we have shown that there exists a basis $\{\tilde{f}_i\}_{i=0}^{m-1}$ of $\mathbb{F}_{\mathcal{A}}[x]$ such that the m polynomials $\tilde{f}_i$ have distinct $d_0 < d_1 < \dots < d_{m-1}$, with $d_0 = 0$ and $d_{m-1} < |A|$. Every polynomial in $\mathbb{F}_{\mathcal{A}}[x]$ is a linear combination of these $\tilde{f}_i$, and therefore has degree d_i for some $i \in \{0, \dots, m-1\}$. $\square$

At the beginning of this section, we took $A \subseteq \mathbb{F}$, the set that generates the $\mathbb{F}$ -algebra $\mathbb{F}_{\mathcal{A}}[x]$, to be an arbitrary set partitioned into m disjoint subsets, without specifying the size of these subsets. In the special case where A has size n , and the partition $\mathcal{A} = \{A_1, \dots, A_m\}$ satisfies

$$|A_i| = r + 1 \quad \text{for all } i = 1 \dots m,$$

we observe the following.

Corollary 5.3. *Assume that the set of degrees $\{d_0, d_1, \dots, d_{m-1}\}$ of the polynomials in $\mathbb{F}_{\mathcal{A}}[x]$ described in Proposition 5.2 (part (4)) includes $d_1 = r + 1$. That is, suppose there exists a polynomial $g \in \mathbb{F}_{\mathcal{A}}[x]$ of degree exactly $r + 1$. Then,*

$$d_j = j(r + 1) \quad \text{for all } j = 0, 1, \dots, m - 1,$$

and the set of polynomials

$$\{1, g, g^2, \dots, g^{m-1}\}$$

defined in Proposition 5.2 (part (3)) forms a basis for $\mathbb{F}_{\mathcal{A}}[x]$.

This result shows that, under a uniform partition where each $A_i \in \mathcal{A}$ satisfies $|A_i| = r + 1$, the algebra $\mathbb{F}_{\mathcal{A}}[x]$ is generated by a single polynomial g of degree $r + 1$, and all other basis elements can be obtained as successive powers of g .

Proof. We show that if such a polynomial g of degree $r + 1$ exists, then it evaluates to a distinct constant on each subset $A_i \in \mathcal{A}$.

Aiming for contradiction, suppose that $g(A_i) = g(A_k)$ for some $i \neq k$. Then the polynomial $g - c$, where $c = g(A_i) = g(A_k)$, vanishes on the union $A_i \cup A_k$, which contains $2(r + 1)$ distinct points. Since $\deg(g - c) = \deg(g) = r + 1$ by construction, this leads to a contradiction.

Hence, g takes distinct values on the m blocks of the partition $\mathcal{A}$. Clearly, $\deg(g) = d_1 = r + 1 < |A| = n$. It then follows from Proposition 5.2 (part (3)) that the polynomials $\{1, g, g^2, \dots, g^{m-1}\}$ form a basis of the $\mathbb{F}$ -algebra $\mathbb{F}_{\mathcal{A}}[x]$.

Moreover, since $\deg(g) = r + 1$, the degree of the monomials g^j are given by $\deg(g^j) = j(r + 1)$, for $j = 0, \dots, m - 1$. $\square$

Note that **Construction 1** relies on the existence of a good polynomial of degree $r + 1$. In other words, the $\mathbb{F}$ -algebra $\mathbb{F}_{\mathcal{A}}[x]$ in **Construction 1** contains such a polynomial. This good polynomial satisfies the assumptions of Corollary 5.3, and therefore the $\mathbb{F}$ -algebra $\mathbb{F}_{\mathcal{A}}[x]$ in **Construction 1** is generated by its powers.

5.2 Construction 2

Having defined an $\mathbb{F}$ -algebra of polynomials associated to the partition $\mathcal{A}$ of a set $A \subseteq F$, we now proceed to utilize the structural properties this $\mathbb{F}$ -algebra to describe the second construction of (n, k, r) LRC codes over finite fields.

Let $A \subseteq \mathbb{F}$ with $|A| = n$, and assume that A is partitioned into $m = \frac{n}{r+1}$ disjoint subsets, each of size $r + 1$. We denote this partition by $\mathcal{A}$:

$$\mathcal{A} = \{A_1, \dots, A_m\}, \quad |A_j| = r + 1 \text{ for all } j = 1, \dots, m.$$

Define the following space of polynomials:

$$\mathcal{F}_{\mathcal{A}}^r := \bigoplus_{i=0}^{r-1} \mathbb{F}_{\mathcal{A}}[x]x^i,$$

where $\mathbb{F}_{\mathcal{A}}[x]$ is the space of polynomials introduced in (17), and the factor x^i serves to shift the polynomials in the i^{th} component. That is, $\mathcal{F}_{\mathcal{A}}^r$ is the direct sum of r independent copies of the $\mathbb{F}$ -algebra $\mathbb{F}_{\mathcal{A}}[x]$, each shifted by multiplication with a power of x . Since each space $\mathbb{F}_{\mathcal{A}}[x]x^i$ has dimension m , it follows that $\dim(\mathcal{F}_{\mathcal{A}}^r) = mr$.

Let $\Phi : \mathbb{F}^k \hookrightarrow \mathcal{F}_{\mathcal{A}}^r$ be an injective mapping. Note that such an injective mapping exists if and only if $k \leq \dim(\mathcal{F}_{\mathcal{A}}^r) = mr = \frac{n}{r+1} \cdot r = \frac{nr}{r+1}$ as this ensures that the image of $\mathbb{F}^k$ can be embedded in $\mathcal{F}_{\mathcal{A}}^r$.

The mapping Φ sends the set of message vectors in $\mathbb{F}^k$ to a set of encoding polynomials in $\mathcal{F}_{\mathcal{A}}^r$, by assigning to each message vector $\mathbf{a} \in \mathbb{F}^k$ a unique corresponding encoding polynomial $f_{\mathbf{a}}(x) := \Phi(\mathbf{a}) \in \mathcal{F}_{\mathcal{A}}^r$. To construct an (n, k, r) LRC code, we evaluate each polynomial $f_{\mathbf{a}}(x) \in \Phi(\mathbb{F}^k)$ at all n distinct points of the evaluation set A . If Φ is a linear mapping, the resulting code is a linear (n, k, r) LRC code.

In order for distinct message vectors to yield distinct codewords, the evaluation map must be injective. This injectivity is guaranteed if all encoding polynomials have degree strictly less than n ; that is, if $\Phi(\mathbb{F}^k) \subseteq \mathbb{F}[x]_{\deg < n}$. Indeed, if $f_{\mathbf{a}}, f_{\mathbf{b}} \in \Phi(\mathbb{F}^k)$ are distinct and $\deg(f_{\mathbf{a}} - f_{\mathbf{b}}) < n$, then the difference $f_{\mathbf{a}} - f_{\mathbf{b}}$ is a nonzero polynomial of degree less than n , and hence cannot vanish at all n distinct evaluation points — since such a polynomial can have at most $n - 1$ distinct roots. In contrast, if $\deg(f_{\mathbf{a}} - f_{\mathbf{b}}) \geq n$, the difference may vanish at every point in the evaluation set A , potentially resulting in $f_{\mathbf{a}}(\alpha) = f_{\mathbf{b}}(\alpha)$ for all $\alpha \in A$ despite $f_{\mathbf{a}} \neq f_{\mathbf{b}}$.

This new construction generalizes **Construction 1** by relaxing the structural constraints imposed on the encoding polynomials used therein. Specifically, **Construction 1** employs a particular linear mapping of the form

$$\mathbf{a} \mapsto \sum_{i=0}^{r-1} f_i(x) x^i$$

where the coefficient polynomials f_i are linear combinations of powers of a fixed good polynomial $g(x)$. This reliance on a fixed good polynomial imposes rigid algebraic structure on the partition $\mathcal{A}$, as the construction depends on the $\mathbb{F}$ -algebra generated by $g(x)$ (i.e., it restricts admissible partition to special forms, such as cosets of subgroups). In contrast, the present construction allows the mapping Φ from message vectors to encoding polynomials to be arbitrary (linear or nonlinear), provided it is injective. Consequently, this approach removes the dependence on a fixed good polynomial and enable a broader range of choices for the coefficient polynomials — allowing to use any partition $\mathcal{A}$ and any basis of $\mathbb{F}_{\mathcal{A}}[x]$. Therefore, **Construction 2** offers a more general and flexible framework for constructing LRC codes.

The following theorem establishes that **Construction 2** indeed yields LRC codes and discusses their minimum distance. Throughout, we denote the encoding polynomial corresponding to the message vector $\mathbf{a}$ by $f_{\mathbf{a}}(x) := \Phi(\mathbf{a})$.

Theorem 5.4. *Construction 2 yields an (n, k, r) LRC code with minimum distance d satisfying the bound*

$$d \geq n - \max_{\mathbf{a}, \mathbf{b} \in \mathbb{F}^k} \deg(f_{\mathbf{a}} - f_{\mathbf{b}}) \geq n - \max_{\mathbf{a} \in \mathbb{F}^k} \deg(f_{\mathbf{a}}). \quad (20)$$

Proof. 1. *Locality property:* We follow the approach used in the proof of Theorem 4.2 For a given message vector $\mathbf{a} \in \mathbb{F}^k$ let the encoding poly $f_{\mathbf{a}}(x)$ by constructed as follows:

$$f_{\mathbf{a}}(x) = \sum_{i=0}^{r-1} f_i(x) x^i \in \mathcal{F}_{\mathcal{A}}^r, \quad (21)$$

where each coefficient polynomial $f_i(x) \in \mathbb{F}_{\mathcal{A}}[x]$.

Fix $j \in \{1, \dots, m\}$, and suppose the symbol to be recovered is $f_{\mathbf{a}}(\alpha)$ for some $\alpha \in A_j$. Define the decoding polynomial

$$\delta_{\alpha}(x) = \sum_{i=0}^{r-1} f_i(\alpha) x^i. \quad (22)$$

Using (21) and (22), we obtain:

$$\delta_{\alpha}(\alpha) = \sum_{i=0}^{r-1} f_i(\alpha) \alpha^i = f_{\mathbf{a}}(\alpha).$$

Furthermore, since each $f_i \in \mathbb{F}_{\mathcal{A}}[x]$, it is constant on each subset $A_j \in \mathcal{A}$. In particular, for all $\beta \in A_j \setminus \{\alpha\}$, we have $f_i(\beta) = f_i(\alpha)$. Thus,

$$f_{\mathbf{a}}(\beta) = \sum_{i=0}^{r-1} f_i(\beta) \beta^i = \sum_{i=0}^{r-1} f_i(\alpha) \beta^i = \delta_{\alpha}(\beta), \quad \text{for all } \beta \in A_j.$$

Since the polynomial $\delta_{\alpha}(x)$ has degree at most $r - 1$, it can be uniquely interpolated using the r values of $f_{\mathbf{a}}(\beta) = \delta_{\alpha}(\beta)$ corresponding to $\beta \in A_j \setminus \{\alpha\}$. Therefore, the erased symbol $f_{\mathbf{a}}(\alpha)$ can be recovered from the remaining r symbols $\{f_{\mathbf{a}}(\beta)\}_{\beta \in A_j \setminus \{\alpha\}}$ within the same subset

A_j of the partition $\mathcal{A}$.

2. *Minimum distance:* Let $(f_{\mathbf{a}}(\alpha))_{\alpha \in A}$ and $(f_{\mathbf{b}}(\alpha))_{\alpha \in A}$ be two codewords constructed from distinct message vectors $\mathbf{a}$ and $\mathbf{b}$. Since the map $\Phi : \mathbb{F}^k \hookrightarrow \mathcal{F}_{\mathcal{A}}^r$ sending $\mathbf{a} \mapsto f_{\mathbf{a}}(x)$ is injective, and $\Phi(\mathbb{F}^k) \subseteq \mathbb{F}[x]_{\deg < n}$, so we have $\deg(f_{\mathbf{a}} - f_{\mathbf{b}}) < n$, the evaluation vectors $(f_{\mathbf{a}}(\alpha))_{\alpha \in A}$ and $(f_{\mathbf{b}}(\alpha))_{\alpha \in A}$ are distinct, as already discussed above.

Let $\mathbf{c}_{\mathbf{a}} := (f_{\mathbf{a}}(\alpha))_{\alpha \in A}$ and $\mathbf{c}_{\mathbf{b}} := (f_{\mathbf{b}}(\alpha))_{\alpha \in A}$. The difference $\mathbf{c}_{\mathbf{a}} - \mathbf{c}_{\mathbf{b}}$ has at most $\deg(f_{\mathbf{a}} - f_{\mathbf{b}})$ zeros among the n evaluation points, so Hamming weight of $\mathbf{c}_{\mathbf{a}} - \mathbf{c}_{\mathbf{b}}$ satisfies

$$w(\mathbf{c}_{\mathbf{a}} - \mathbf{c}_{\mathbf{b}}) \geq n - \deg(f_{\mathbf{a}} - f_{\mathbf{b}})$$

By Lemma 2.6,

$$d(\mathbf{c}_{\mathbf{a}}, \mathbf{c}_{\mathbf{b}}) = w(\mathbf{c}_{\mathbf{a}} - \mathbf{c}_{\mathbf{b}}),$$

and by Definition 2.7 the minimum distance of the code is

$$d \geq n - \max_{\mathbf{a}, \mathbf{b} \in \mathbb{F}^k} \deg(f_{\mathbf{a}} - f_{\mathbf{b}}).$$

Since subtracting two polynomials can cancel leading terms but cannot increase the degree, we have

$$\max_{\mathbf{a}, \mathbf{b} \in \mathbb{F}^k} \deg(f_{\mathbf{a}} - f_{\mathbf{b}}) \leq \max_{\mathbf{a} \in \mathbb{F}^k} \deg(f_{\mathbf{a}}),$$

and (20) follows. □

6 Systematic encoding of Construction 1

In practical implementations, it is often desirable to encode messages systematically, meaning that the original message symbols are directly embedded in the codeword. This facilitates immediate access to the stored information without decoding. All the LRC code constructions discussed earlier can be modified to produce systematic codes without sacrificing code locality or minimum distance by modifying the encoding polynomials in (6) and (21).

Below, we describe a systematic variant of **Construction 1** for obtaining optimal linear LRC codes, as discussed in Section III.D of [12]. Having seen that **Construction 2** generalizes **Construction 1**, we note that systematic encoding leverages the $\mathbb{F}$ -algebra introduced in **Construction 2** to ensure that the message symbols are directly embedded in the codeword. In particular, the systematic encoding method is presented as a modification of **Construction 1**, using the algebraic framework of **Construction 2**.

We consider the setting where the dimension k is divisible by the locality parameter r , and the code length n is divisible by $r + 1$. As in Section 5.2, let $\mathbb{F}$ denote a finite field. Let $\mathcal{A} = \{A_1, \dots, A_m\}$ be a partition of the set $A \subseteq \mathbb{F}$ of size n into $m = \frac{n}{r+1}$ disjoint subsets, each of size $r + 1$.

Take $i = 1, \dots, \frac{k}{r}$ (note that $\frac{k}{r} \leq \frac{n}{r+1}$ by (3)), and let $B_i = \{\beta_{i,1}, \dots, \beta_{i,r}\}$ be a subset of A_i of size r . The k message symbols will be placed in the codeword positions corresponding to the r locations in the $\frac{k}{r}$ sets B_i .

Recall from Proposition 5.2 (part (2)) in Section 5.2 that the $\mathbb{F}$ -algebra $\mathbb{F}_{\mathcal{A}}[x]$ has a basis $\{f_i\}_{i=1}^m$ of polynomials satisfying

$$f_i(A_j) = \delta_{i,j} \quad \text{for } i, j = 1, \dots, m,$$

where $\delta_{i,j}$ is the Kronecker delta.

For each $B_i = \{\beta_{i,1}, \dots, \beta_{i,r}\}$, with $i \in \{1, \dots, \frac{k}{r}\}$, define r polynomials $\phi_{i,j}(x)$, for $j = 1, \dots, r$, each of degree less than r , such that

$$\phi_{i,j}(\beta_{i,l}) = \delta_{j,l}.$$

These polynomials can be constructed via Lagrange interpolation:

$$\phi_{i,j}(x) = \prod_{\substack{1 \leq k \leq r \\ k \neq j}} \frac{x - \beta_{i,k}}{\beta_{i,j} - \beta_{i,k}},$$

which ensures that $\phi_{i,j}(\beta_{i,j}) = 1$ and $\phi_{i,j}(\beta_{i,k}) = 0$ for $k \neq j$ (i.e., for all other points in $B_i \setminus \{\beta_{i,j}\}$).

Let $\mathbf{a} = (a_{i,j})$ with $i = 1, \dots, \frac{k}{r}$, $j = 1, \dots, r$, be a message vector consisting of k message symbols. Define the encoding polynomial

$$f_{\mathbf{a}}(x) = \sum_{i=1}^{\frac{k}{r}} f_i(x) \left(\sum_{j=1}^r a_{i,j} \phi_{i,j}(x) \right). \quad (23)$$

The corresponding codeword is obtained by evaluating $f_{\mathbf{a}}(x)$ at all points $a \in A$, resulting in the vector $(f_{\mathbf{a}}(a))_{a \in A}$, as in (7) of **Construction 1**.

The encoding polynomial $f_{\mathbf{a}}(x)$ defined in (23) belongs to $\mathcal{F}_{\mathcal{A}}^r$, and hence, by Theorem 5.4, the code has locality r . To see this, observe that for each i ,

$$\psi_i(x) := \sum_{j=1}^r a_{i,j} \phi_{i,j}(x)$$

is a polynomial of degree less than r , since it is a linear combination of the polynomials $\phi_{i,j}(x)$, each of which has degree less than r , as seen earlier. Specifically, each $\phi_{i,j}(x)$ is obtained via polynomial interpolation over the r locations in B_i . Each $a_{i,j} \in \mathbb{F}$, and the same holds for the coefficients of each $\phi_{i,j}(x)$, so we may write $\psi_i(x)$ as

$$\psi_i(x) = \sum_{t=0}^{r-1} b_{i,t} x^t \quad \text{for some } b_{i,t} \in \mathbb{F},$$

with $\deg(\psi_i(x)) < r$. Thus, for each i ,

$$f_i(x) \psi_i(x) = f_i(x) \left(\sum_{t=0}^{r-1} b_{i,t} x^t \right) = \sum_{t=0}^{r-1} (b_{i,t} f_i(x)) x^t.$$

Note that $b_{i,t} f_i(x) \in \mathbb{F}_{\mathcal{A}}[x]$, since $\mathbb{F}_{\mathcal{A}}[x]$ is closed under scalar multiplication. Then,

$$f_{\mathbf{a}}(x) = \sum_{i=1}^{\frac{k}{r}} f_i(x) \psi_i(x) = \sum_{i=1}^{\frac{k}{r}} \sum_{t=0}^{r-1} (b_{i,t} f_i(x)) x^t = \sum_{t=0}^{r-1} \left(\sum_{i=1}^{\frac{k}{r}} (b_{i,t} f_i(x)) \right) x^t.$$

Since $\mathbb{F}_{\mathcal{A}}[x]$ is closed under addition, it follows that each $\sum_{i=1}^{\frac{k}{r}} (b_{i,t} f_i(x)) \in \mathbb{F}_{\mathcal{A}}[x]$, and hence $f_{\mathbf{a}}(x) \in \mathbb{F}_{\mathcal{A}}^r$.

Moreover, observe that, by definition, for a fixed $\beta_{i,j}$, we have $f_i(\beta_{i,j}) = 1$, while $f_l(\beta_{i,j}) = 0$ for all $l \neq i$.

Similarly, $\phi_{i,j}(\beta_{i,j}) = 1$, whereas $\phi_{i,k}(\beta_{i,j}) = 0$ for all $k \neq j$. It follows that

$$f_{\mathbf{a}}(\beta_{i,j}) = a_{i,j}, \quad \text{for all } i = 1, \dots, \frac{k}{r} \text{ and } j = 1, \dots, r,$$

so the code is indeed systematic.

Before analyzing the optimality of the LRC codes constructed above, we present an example to illustrate the underlying construction principles.

Example 6.1. Let $\mathbb{F} = \mathbb{F}_7$ be our finite field and consider the parameters $(n, k, r) = (6, 4, 2)$.

Since $r = 2$, we have $\frac{n}{r+1} = \frac{6}{3} = 2$, so we take a partition consisting of two disjoint subsets of size $r + 1 = 3$ as follows:

$$A_1 = \{1, 2, 4\}, \quad A_2 = \{3, 6, 5\},$$

where A_1 is the order 3 subgroup of $\mathbb{F}_7^\times$ generated by 2, and A_2 is its coset $3A_1$. The evaluation set A is therefore $A = \{1, 2, 3, 4, 5, 6\} \subset \mathbb{F}_7$ of size $n = 6$

For $i = 1, 2$, select a subset $B_i \subset A_i$ of size $r = 2$:

$$B_1 = \{\beta_{1,1} = 1, \beta_{1,2} = 2\}, \quad B_2 = \{\beta_{2,1} = 3, \beta_{2,2} = 5\}.$$

Thus, the $k = 4$ message symbols will be encoded at locations 1, 2, 3, 5, while locations 4 and 6 will be reserved for redundancy symbols. For $j = 1, 2$, let the message vector be

$$\mathbf{a} = (a_{1,1}, a_{1,2}, a_{2,1}, a_{2,2}) = (3, 0, 2, 1).$$

For each set B_i , define functions $\phi_{i,j}(x)$ such that

$$\phi_{i,j}(\beta_{i,l}) = \delta_{j,l}$$

using Lagrange interpolation over B_i . Working in $\mathbb{F}_7$, we proceed as follows.

In $B_1 = \{1, 2\}$:

- $\phi_{1,1}(x) = 1$ satisfies $\phi_{1,1}(1) = 1, \phi_{1,1}(2) = 0$:

$$\phi_{1,1}(x) = \frac{x-2}{1-2} = -(x-2) = 6(x-2),$$

- $\phi_{1,2}(x)$ satisfies $\phi_{1,2}(2) = 1, \phi_{1,2}(1) = 0$:

$$\phi_{1,2}(x) = \frac{x-1}{2-1} = x-1.$$

In $B_2 = \{3, 5\}$:

- $\phi_{2,1}(x)$ satisfies $\phi_{2,1}(3) = 1, \phi_{2,1}(5) = 0$:

$$\phi_{2,1}(x) = \frac{x-5}{3-5} = \frac{x-5}{-2} = 3(x-5),$$

- $\phi_{2,2}(x)$ satisfies $\phi_{2,2}(5) = 1, \phi_{2,2}(3) = 0$:

$$\phi_{2,2}(x) = \frac{x-3}{5-3} = 4(x-3).$$

To construct the encoding polynomial as defined in (23), we compute the inner sum

$$\sum_{j=1}^2 a_{i,j} \phi_{i,j}(x)$$

for each $i = 1, 2$.

For $i = 1$, this yields:

$$a_{1,1} \phi_{1,1}(x) + a_{1,2} \phi_{1,2}(x) = 3 \cdot 6(x-2) + 0 \cdot (x-1) = 4(x-2).$$

For $i = 2$, we obtain:

$$a_{2,1} \phi_{2,1}(x) + a_{2,2} \phi_{2,2}(x) = 2 \cdot 3(x-5) + 1 \cdot 4(x-3) = 3x.$$

Substituting these expressions into (23) gives:

$$f_{\mathbf{a}}(x) = f_1(x) \cdot 4(x-2) + f_2(x) \cdot 3x,$$

where $f_1(x)$ and $f_2(x)$ are the basis polynomials of $\mathbb{F}_{\mathcal{A}}[x]$ that satisfy:

$$f_i(A_j) = \delta_{i,j} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{otherwise,} \end{cases} \quad \text{for } i, j = 1, 2.$$

For completeness, we evaluate $f_{\mathbf{a}}(x)$ at the location in B_1 and B_2 in order to verify that these evaluations indeed correspond to the message symbols.

For $B_1 = \{1, 2\}$ we have:

$$\begin{aligned} f_{\mathbf{a}}(\beta_{1,1}) &= f_{\mathbf{a}}(1) = 1 \cdot -4 + 0 = 3 = a_{1,1}, \\ f_{\mathbf{a}}(\beta_{1,2}) &= f_{\mathbf{a}}(2) = 1 \cdot 0 + 0 = 0 = a_{1,2}. \end{aligned}$$

For $B_2 = \{3, 5\}$ we find:

$$\begin{aligned} f_{\mathbf{a}}(\beta_{2,1}) &= f_{\mathbf{a}}(3) = 0 + 1 \cdot 2 = 2 = a_{2,1}, \\ f_{\mathbf{a}}(\beta_{2,2}) &= f_{\mathbf{a}}(5) = 0 + 1 \cdot 1 = 1 = a_{2,2}. \end{aligned}$$

It follows that the corresponding codeword is given by:

$$(f_{\mathbf{a}}(1), f_{\mathbf{a}}(2), f_{\mathbf{a}}(4), f_{\mathbf{a}}(3), f_{\mathbf{a}}(6), f_{\mathbf{a}}(5)) = (3, 0, \text{redundancy}_1, 2, \text{redundancy}_2, 1).$$

To determine the remaining redundancy symbols, we evaluate $f_{\mathbf{a}}$ at $x = 4$ and $x = 6$, respectively:

$$\begin{aligned} f_{\mathbf{a}}(4) &= 1 \cdot 1 + 0 = 1, \\ f_{\mathbf{a}}(6) &= 0 + 1 \cdot 4 = 4. \end{aligned}$$

Hence, the complete codeword becomes:

$$(3, 0, 1, 2, 4, 1).$$

We see that the original message symbols 3, 0, 2, 1 appear directly in the codeword, confirming that the code is systematic.

In the preceding example, we have $\frac{k}{r} = \frac{n}{r+1}$, so the index i ranges over all the subsets in the partition of

the evaluation set. In contrast, the following example demonstrates the case where $\frac{k}{r} \leq \frac{n}{r+1}$, so only a proper subset of the partition is involved in encoding the message symbols systematically.

Example 6.2. Let us construct a systematic $(12, 4, 2)$ LRC code over $\mathbb{F}_{13}$.

Let

$$\mathcal{A} = \left\{ A_1 = \{1, 3, 9\}, A_2 = \{2, 6, 5\}, A_3 = \{4, 12, 10\}, A_4 = \{7, 8, 11\} \right\}.$$

be a partition of $A = \mathbb{F}_{13}^\times \subset \mathbb{F}_{13}$ of size $n = 12$ into $\frac{n}{r+1} = \frac{12}{3} = 4$ subsets, each of size $r + 1 = 3$.

For $i = 1, 2$, let $B_i \subset A_i$ be:

$$B_1 = \{\beta_{1,1} = 1, \beta_{1,2} = 3\}, \quad B_2 = \{\beta_{2,1} = 2, \beta_{2,2} = 6\}.$$

Thus, $k = 4$ message symbols will be encoded at locations 1, 3, 2, 6, while the evaluations at the remaining location in A will store redundancy symbols. Let the message vector be

$$\mathbf{a} = (a_{1,1}, a_{1,2}, a_{2,1}, a_{2,2}) = (3, 0, 2, 1).$$

Define the Lagrange basis polynomials $\phi_{i,j}(x)$ on each set B_i , for $i = 1, 2$ and $j = 1, 2$, satisfying $\phi_{i,j}(\beta_{i,l}) = \delta_{j,l}$:

$$\begin{aligned} \phi_{1,1}(x) &= \frac{x-3}{1-3} = \frac{x-3}{-2} = 6(x-3) = 6x+8, \\ \phi_{1,2}(x) &= \frac{x-1}{3-1} = \frac{x-1}{2} = 7(x-1) = 7x+6, \\ \phi_{2,1}(x) &= \frac{x-6}{2-6} = \frac{x-6}{-4} = 3(x-6) = 3x+8, \\ \phi_{2,2}(x) &= \frac{x-2}{6-2} = \frac{x-2}{4} = 10(x-2) = 10x+6. \end{aligned}$$

Computing $\sum_{j=1}^2 a_{i,j} \phi_{i,j}(x)$ for each $i = 1, 2$ we have:

- For $i = 1$,

$$a_{1,1} \phi_{1,1}(x) + a_{1,2} \phi_{1,2}(x) = 3(6x+8) + 0(7x+6) = 5x+11.$$

- For $i = 2$,

$$a_{2,1} \phi_{2,1}(x) + a_{2,2} \phi_{2,2}(x) = 2(3x+8) + 1(10x+6) = 3x+9.$$

The full encoding polynomial becomes:

$$f_{\mathbf{a}}(x) = f_1(x) \cdot (5x+11) + f_2(x) \cdot (3x+9),$$

where $f_1(x)$ and $f_2(x)$ are two of the basis polynomials of $\mathbb{F}_{\mathcal{A}}[x]$ that satisfy:

$$f_i(A_j) = \delta_{i,j} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{otherwise,} \end{cases} \quad \text{for all } i, j = 1, \dots, 4.$$

Since the encoding polynomial only involves $f_1(x)$, $f_2(x)$, which are supported on A_1 and A_2 respectively, it follows that

$$f_1(A_3) = f_1(A_4) = f_2(A_3) = f_2(A_4) = 0.$$

By construction,

$$\begin{aligned} f_{\mathbf{a}}(\beta_{1,1}) &= f_{\mathbf{a}}(1) = 3 = a_{1,1}, \\ f_{\mathbf{a}}(\beta_{1,2}) &= f_{\mathbf{a}}(3) = 0 = a_{1,2}, \\ f_{\mathbf{a}}(\beta_{2,1}) &= f_{\mathbf{a}}(2) = 2 = a_{2,1}, \\ f_{\mathbf{a}}(\beta_{2,2}) &= f_{\mathbf{a}}(6) = 1 = a_{2,2}. \end{aligned}$$

Furthermore, evaluating $f_{\mathbf{a}}(x)$ at the remaining locations in $A \setminus (B_1 \cup B_2)$ yields:

$$\begin{aligned} f_{\mathbf{a}}(9) &= 1 \cdot 4 + 0 = 4, \\ f_{\mathbf{a}}(5) &= 0 + 1 \cdot 11 = 11, \\ f_{\mathbf{a}}(A_3) &= 0, \\ f_{\mathbf{a}}(A_4) &= 0. \end{aligned}$$

Therefore, the final codeword is:

$$\begin{aligned} \mathbf{c} &= (f_{\mathbf{a}}(1), f_{\mathbf{a}}(3), f_{\mathbf{a}}(9), f_{\mathbf{a}}(2), f_{\mathbf{a}}(6), f_{\mathbf{a}}(5), f_{\mathbf{a}}(4), f_{\mathbf{a}}(12), f_{\mathbf{a}}(10), f_{\mathbf{a}}(7), f_{\mathbf{a}}(8), f_{\mathbf{a}}(11)) \\ &= (3, 0, 4, 2, 1, 11, 0, 0, 0, 0, 0, 0). \end{aligned}$$

While this construction, using the encoding polynomial in (23), produces a systematic (n, k, r) LRC code, it does not necessarily achieve optimal minimum distance just yet. This limitation arises because $f_{\mathbf{a}}(x) \in \mathbb{F}_{\mathcal{A}}^r$ and its degree, as seen in the proof of Theorem 5.4, is only constrained by the number of distinct evaluation points assigned to it — specifically, $\deg(f_{\mathbf{a}}) < |A| = n$ is the best bound we can guarantee without imposing additional structure.

To address this issue, we invoke Proposition 5.2 (part (3)) and Corollary 5.3. We will see that if the $\mathbb{F}$ -algebra $\mathbb{F}_{\mathcal{A}}[x]$ is generated by the powers of a good polynomial $g(x)$, as is the case for **Construction 1** (where the good polynomial $g(x)$ has degree $r + 1$), then we can construct an optimal systematic LRC code. In particular, the polynomials $f_i(x)$ appearing in (23) can be replaced by polynomials $\widetilde{f}_i(x)$ that are linear combinations of the polynomials $1, g, g^2, \dots, g^{\frac{k}{r}-1}$ (recall that $\{1, g, g^2, \dots, g^{m-1}\}$ forms a basis for $\mathbb{F}_{\mathcal{A}}[x]$, where $m = \frac{n}{r+1}$, and that since $\frac{k}{r} \leq \frac{n}{r+1}$ by (3), the polynomials $1, g, g^2, \dots, g^{\frac{k}{r}-1}$ exist). These $\widetilde{f}_i(x)$ must satisfy:

$$\widetilde{f}_i(A_j) = \delta_{i,j} \quad \text{for } i, j = 1, \dots, \frac{k}{r}. \quad (24)$$

The existence of such polynomials $\widetilde{f}_i(x)$ is ensured by the invertibility of the Vandermonde matrix

$$V = (g^{j-1}(A_i)).$$

Namely, consider the following. Our goal is to construct polynomials $\widetilde{f}_1(x), \dots, \widetilde{f}_{\frac{k}{r}}(x)$ such that:

1. Each $\widetilde{f}_i(x)$ lies in the span of the polynomials $1, g, g^2, \dots, g^{\frac{k}{r}-1}$, and
2. The condition $\widetilde{f}_i(A_j) = \delta_{i,j}$ holds for all $i, j = 1, \dots, \frac{k}{r}$.

Following the strategy used in the proof of Proposition 5.2 (part (3)), we write each $\widetilde{f}_i(x)$ as a linear combination of the powers of $g(x)$, namely:

$$\widetilde{f}_i(x) = \sum_{l=0}^{\frac{k}{r}-1} c_{i,l} g(x)^l \quad (25)$$

so that the condition in (24) becomes a linear system in the coefficient vector $\mathbf{c}_i = (c_{i,0}, \dots, c_{i,\frac{k}{r}-1})^T$. That is,

$$V \mathbf{c}_i = \mathbf{e}_i,$$

where $\mathbf{e}_i \in \mathbb{F}_{\mathcal{A}}^{\frac{k}{r}}$ is the standard unit vector with a 1 in the i^{th} position, and

$$V = (g^{j-1}(A_i))_{1 \leq i, j \leq \frac{k}{r}}$$

is a $\frac{k}{r} \times \frac{k}{r}$ Vandermonde matrix formed from the values $g(A_1), \dots, g(A_{\frac{k}{r}})$. The matrix V is invertible

provided that the values $g(A_1), \dots, g(A_{\frac{k}{r}})$ are distinct, which is guaranteed by the assumption that g is a good polynomial. It follows that the system has a unique solution, thus the polynomials $\widetilde{f}_i(x)$ can be constructed uniquely.

From (25), we see that the degree of $\widetilde{f}_i(x)$ is at most $(\frac{k}{r}-1)(r+1)$, which, together with $\deg(\phi_{i,j}) \leq r-1$, bounds the degree of $f_a(x)$ in (23) by $k + \frac{k}{r} - 2$. As seen in the proof of Theorem 4.2, optimality of the minimum distance follows.

In summary, while the encoding polynomial in (23) yields systematic (n, k, r) LRC codes with locality r , achieving optimality requires using **Construction 1**'s setup, where the $\mathbb{F}$ -algebra $\mathbb{F}_{\mathcal{A}}[x]$ is generated by the powers of a good polynomial of degree $r+1$.

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